

High-Power Inductive Charging System Development and Integration for Mobility

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U.S. DEPARTMENT OF
ENERGY

Overview

Timeline

- Start – October 2018 (FY19)
- End – September 2021 (FY21)
- 3 budget periods

Budget

- Total project funding
 - DOE share – \$2.2M
 - Cost share from partners – \$3.0M
- Project spending BP1: \$850K
- Project spending BP2: \$950K

Barriers

- Achieving 100- and 270-kW charge rates for light-duty passenger vehicles to meet the ~3C charge rates (50% increase in SOC in 10 minutes)
- High operating efficiencies $\geq 90\%$ (end-to-end) for both cases
- Interoperable grid side power electronics and electromagnetics hardware

Partners



- ORNL (Project lead)



- Hyundai-Kia America Technical Center (HATCI)



Project Objectives and Relevance

Overall Objectives:

- Design a high power, modular, scalable, interoperable, high-efficiency plug-less extreme fast charging system
- Have minimal grid level disruptions with $\leq 5\%$ harmonic distortions on the grid current and $\geq 95\%$ grid power factor
- Design and develop a compact and light-weight poly-phase electromagnetic coupling coils that can be scaled up to power transfer levels that correspond to 3C charge rates
- Achieve high charging efficiencies greater than 90% (end-to-end)
- Integrate vehicle to infrastructure charging communication protocols (wireless)
- Understand and address vehicle integration issues of XFC technology, including energy storage impacts and thermal management considerations

FY 2021 Objectives:

- Complete the design and hardware development of the 100/300-kW interoperable primary-side power electronics hardware and transmitter coupler and the 100- and 300-kW rectifier systems and receiver couplers with primary and secondary side resonant tuning networks.
- Integrate system into a Hyundai Kona EV research vehicle provided by HATCI
- Integrate system into a Porsche Taycan EV research vehicle provided by VW Group of America
- Technology demonstration, validation, test and data collection

Relevance:

- Project is aligned with the VTO Grid & Infrastructure Program's objective: *"Identify and assess system requirements and conduct research to enable extreme fast charging while minimizing impacts to the grid."*
- Program Focus: *"Enhance EV refueling with advanced technologies such as wireless power transfer and high-power charging systems"*

Any proposed future work is subject to change based on funding levels

Project Milestones

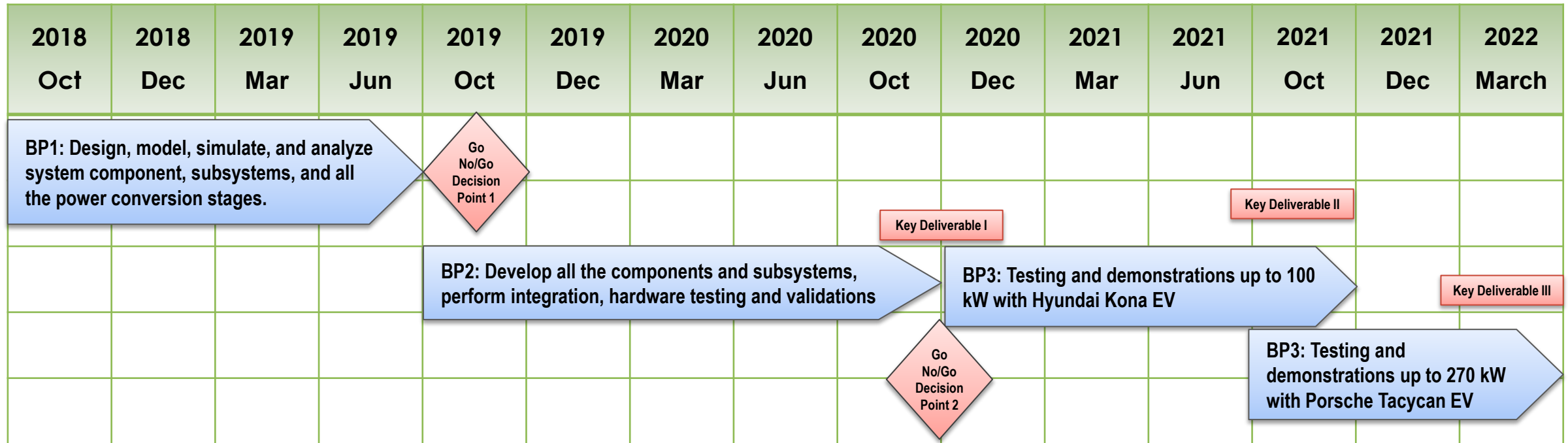
Date	Milestones and Go/No-Go Decisions	Status
BP1 (FY19) 10/2018-09/2019	<i>[Research and design phase]</i> Design, model, simulate, and analyze system components. Size and design all of the subsystems and components.	Completed
BP2 (FY20) 10/2019-09/2020	<i>[System hardware development, integration, and validations]</i> Build all the system components, subsystems, and power conversion stages and integrate with each other.	Completed
BP3 (FY21) 10/2020-09/2021	<i>[System development, integration, and 100- and 270 kW testing and demonstrations]</i> Complete system integration and perform necessary modifications, optimize system design, fine tune components and controllers. Integrate system into a Hyundai Kona EV and Porsche Taycan EV, validate 100- and 270-kW power transfer. Test and collect data and demonstrate the proposed concept.	On track

Any proposed future work is subject to change based on funding levels

Approach / Strategy

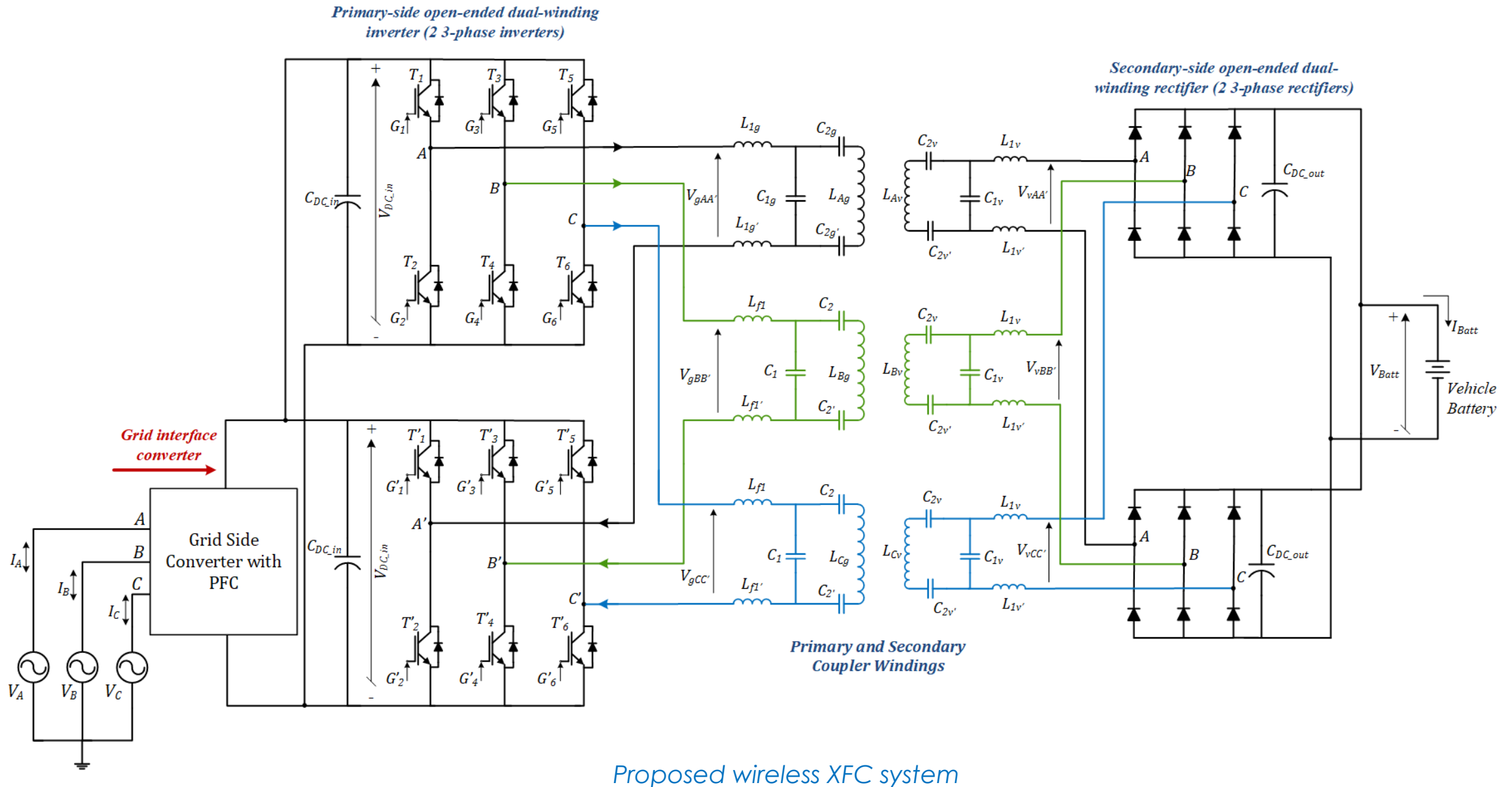
- Iterative design and the use of finite element analysis (FEA) based modeling for the design optimization of the electromagnetic coupling coils
- Target operating conditions (power level, input grid voltage, vehicle battery voltage range, airgap, maximum current, etc.) are used for proper system level design and cascaded down to the appropriate subsystems and component designs
- Model and simulate the grid interface (front-end) power blocks based on the dc link voltage requirements of the proposed system and the grid infrastructure parameters
- Design the system power conversion stages in an integrated approach for an optimal system design in terms of complexity and compactness
- Test and validate all the power conversion stages individually before the full system integration (for functionality and performance)
- Test the entire system using grid and battery emulators before vehicle integration
- Design and develop a prototype for proof of concept before designing and developing the high power scaled couplers and converters
- Complete vehicle integrations, evaluations, and validation of the proposed system

Project Timeline



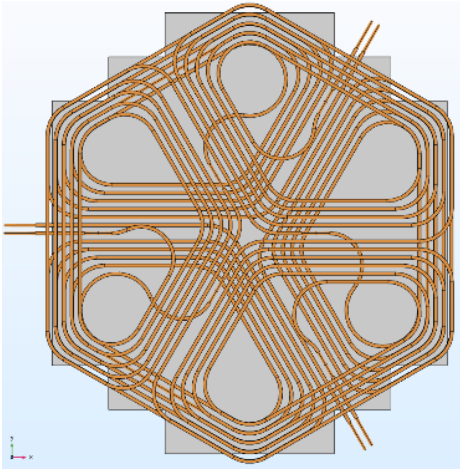
- **Go/No-Go Decision Point 1:** Whether the system design, models, and simulation indicate the feasibility of the proposed XFC system
- **Go/No-Go Decision Point 2:** Completion of hardware development for all the system components and subsystems
- **Key Deliverable I:** Complete design and hardware development of all the power conversion stages
- **Key Deliverable II:** Testing, validations, and demonstrations up to 100-kW power with Hyundai KONA EV
- **Key Deliverable III:** Testing, validations, and demonstrations up to 270-kW power with Porsche Taycan EV

Proposed Technology

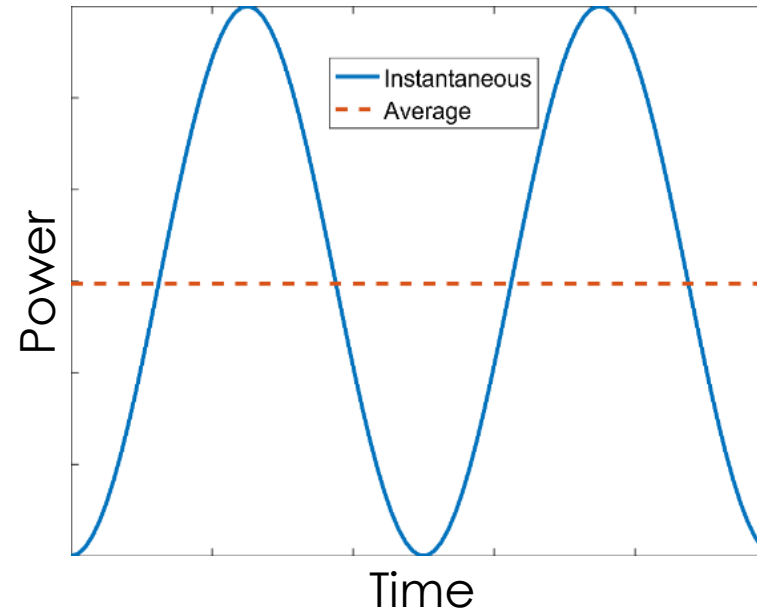


Polyphase Coupler Technology

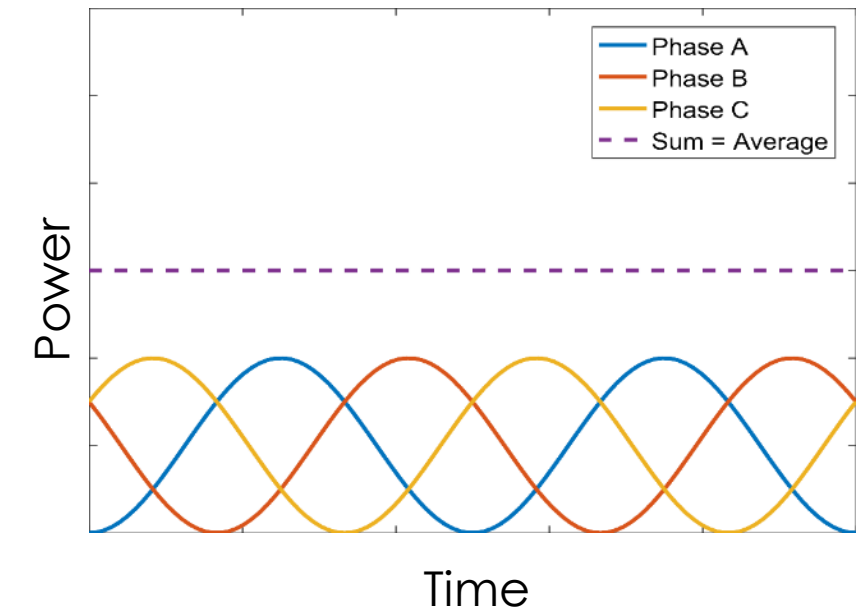
- Single-phase systems “pulse” power across the airgap
- Low space-time average utilization since fields oscillate between peak values and zero



Polyphase coupler wiring



Instantaneous and average power variations for conventional circular couplers



Instantaneous and average power variations for polyphase coupler

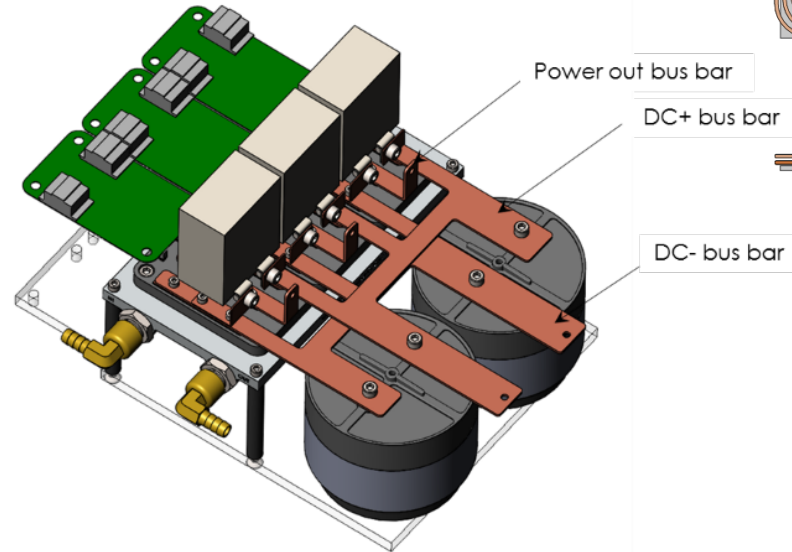
- Polyphase systems use rotating fields for constant power transfer
- Phase-shifted excitation and spatially shifted coils
- Much higher power density due to improved space-time field utilization
- Lower current ripple

Technical Accomplishments – BP I

Designed and prototyped a 3-phase, 2-layer, polyphase couplers

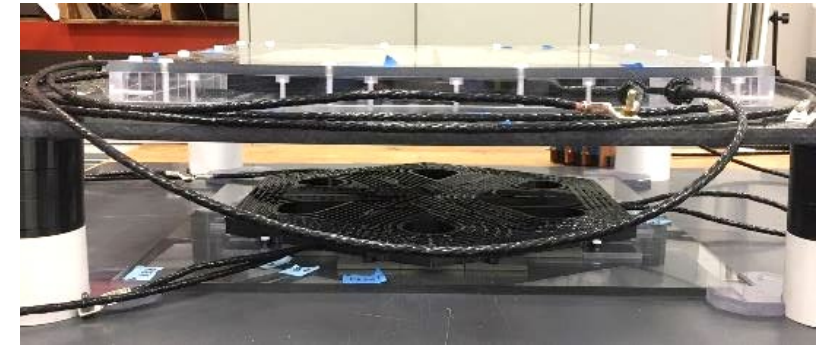
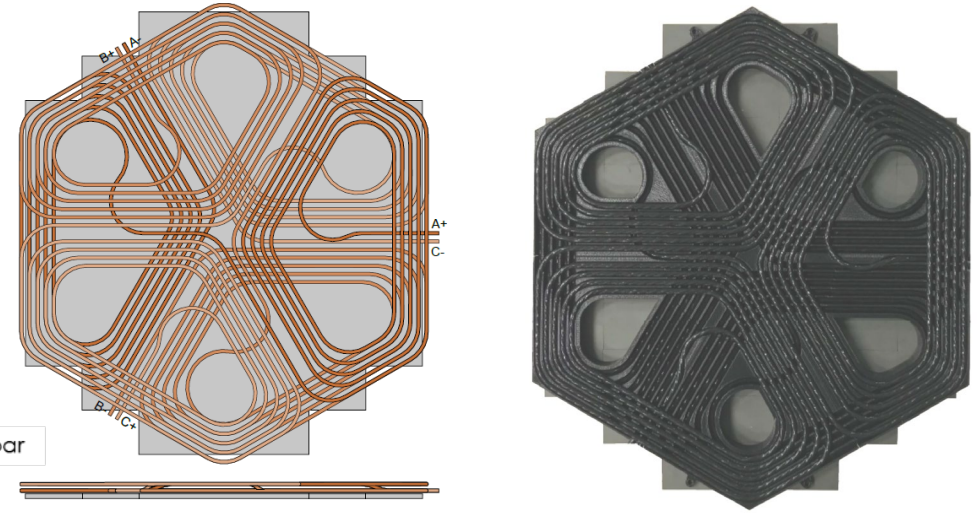
- Polyphase coupler design
- Inductance readings @170mm airgap

		Aligned		$\Delta\varphi=30^\circ$	
		Sim.	Exp.	Sim.	Exp.
GA Self-Inductances	L_a	34.1	31.6	34.1	30.9
	L_b	34.3	32.3	34.3	30.6
	L_c	34.2	32.4	34.2	30.9
	M_{ab}	-11.2	- 9.8	-11.2	- 9.9
	M_{bc}	-11.3	- 9.8	-11.3	-10.2
	M_{ca}	-11.1	-10.2	-11.1	-10.6
VA Self-Inductances	L_a	34.1	31.1	34.1	30.9
	L_b	34.3	31.3	34.3	31.0
	L_c	34.2	31.6	34.2	30.7
	M_{ab}	-11.2	-10.0	-11.2	-10.2
	M_{bc}	-11.3	- 9.8	-11.3	-10.1
	M_{ca}	-11.1	- 9.9	-11.1	- 9.6
GA to VA Mutual-Inductances	M_{ga}^{va}	5.65	5.32	5.32	5.10
	M_{gb}^{va}	-3.64	-3.68	-1.08	-1.49
	M_{gc}^{va}	-1.85	-1.53	-4.29	-3.27
	M_{ga}^{vb}	-1.85	-1.23	-4.29	-3.57
	M_{gb}^{vb}	5.54	5.42	5.20	5.76
	M_{gc}^{vb}	-3.57	-3.50	-0.99	-1.73
	M_{ga}^{vc}	-3.64	-3.32	-1.08	-1.47
	M_{gb}^{vc}	-1.77	-1.30	-4.18	-3.97
	M_{gc}^{vc}	5.55	5.07	5.20	5.40



3-phase inverter design rendered image

- 3-phase inverter design based on ORNL 120 kW WPT system
- Effective self and mutual inductances



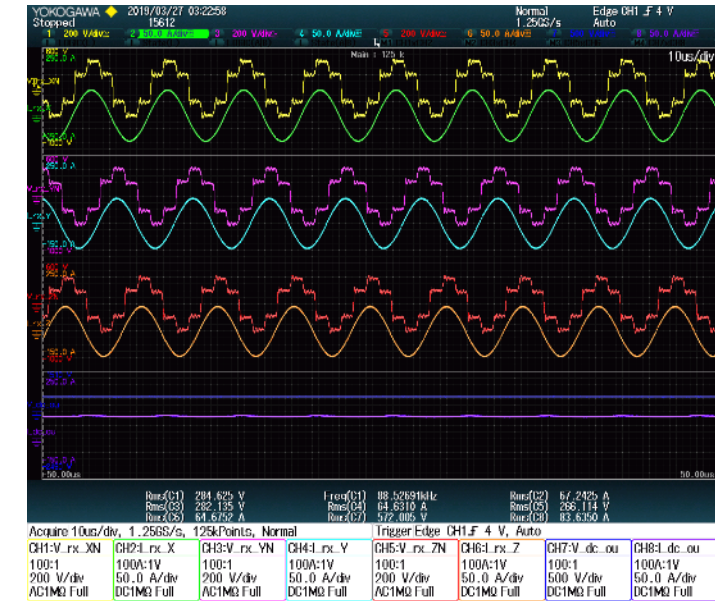
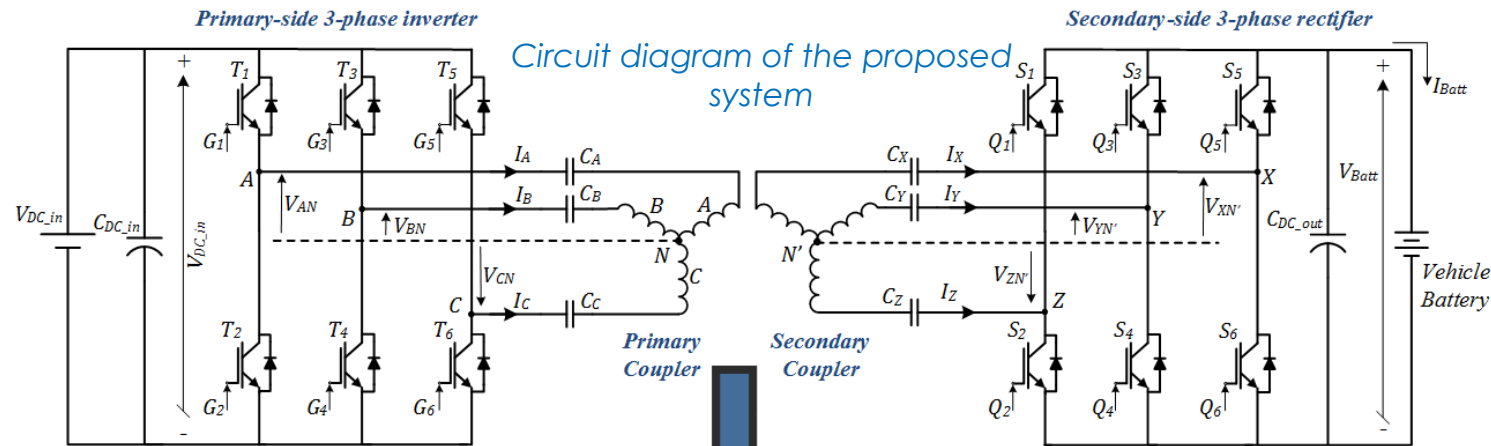
Design and hardware realization of 3-phase polyphase electromagnetic couplers

	Aligned		$\Delta\varphi=30^\circ$		$\Delta x=\Delta y=100\text{ mm}$	
	Sim.	Exp.	Sim.	Exp.	Sim.	Exp.
$L_{g,e}$	136.2	000.0	136.2	123.1	134.7	121.9
$L_{v,e}$	136.2	000.0	136.2	122.5	134.7	121.0
M_e	25.3	00.0	25.1	24.6	12.7	12.7
k_e	0.186	0.000	0.184	0.200	0.094	0.105

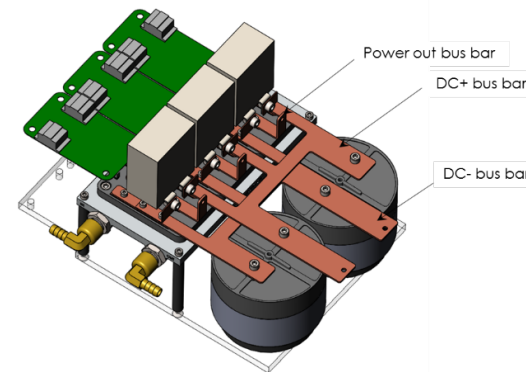
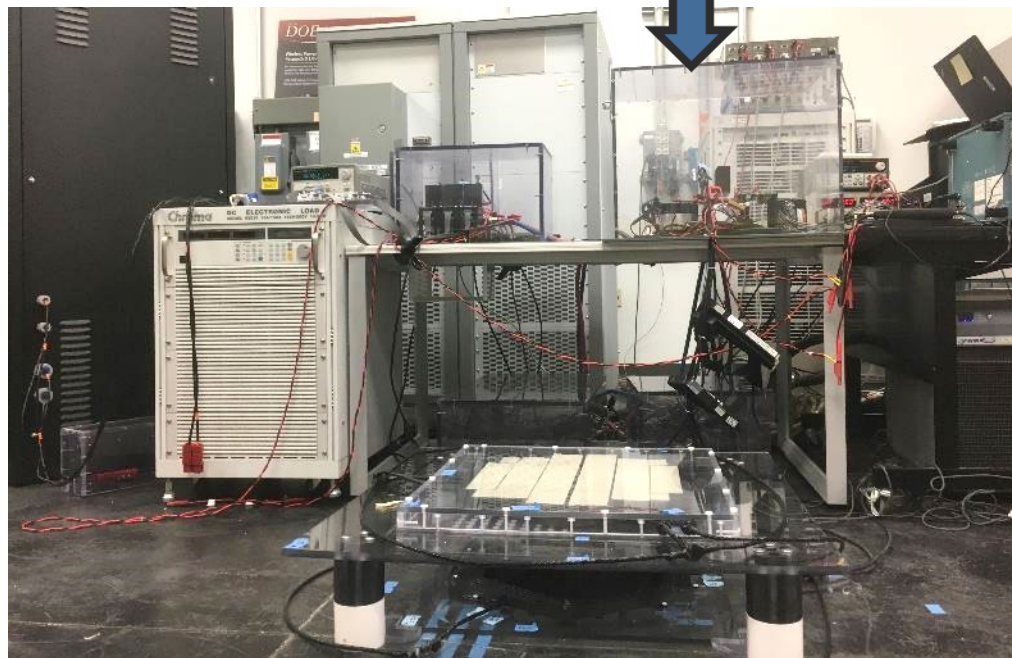
Technical Accomplishments – BP I

Developed a laboratory benchtop test setup to validate the concepts

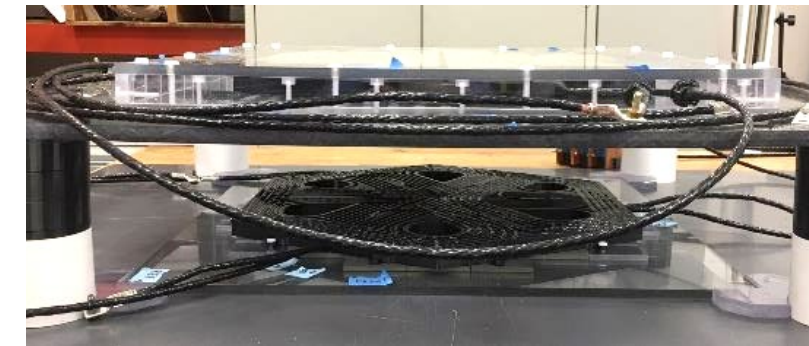
- Hardware development and experimental test results



Secondary-side voltages and currents



3-phase inverter design rendered image

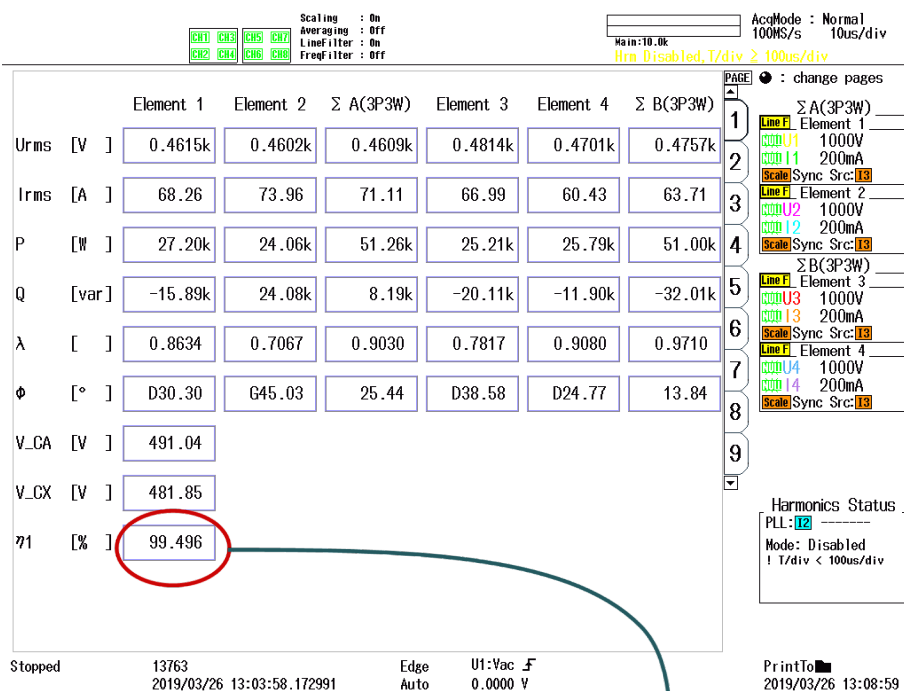


Design and hardware realization of 3-phase polyphase electromagnetic couplers

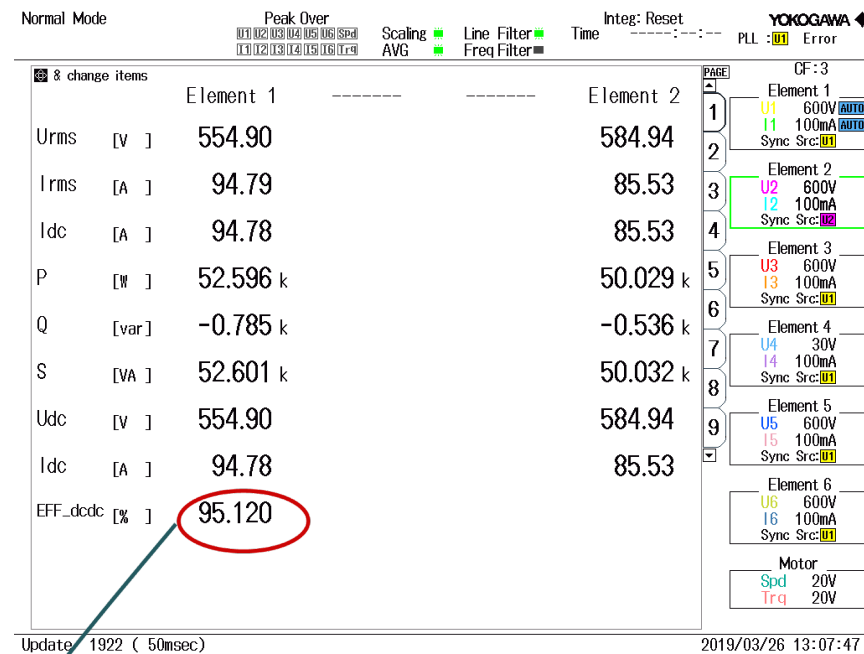
Technical Accomplishments – BP I

Developed a laboratory benchtop test setup to validate the concepts

- Hardware development and experimental test results: 99.49% coil-to-coil efficiency
- Closely matches the simulation results – validating the model



3-phase couplers transmitter to receiver
efficiency analysis with 2-wattmeter
method



55 kW experimental test results (dc input /
inverter input and rectifier output & load)

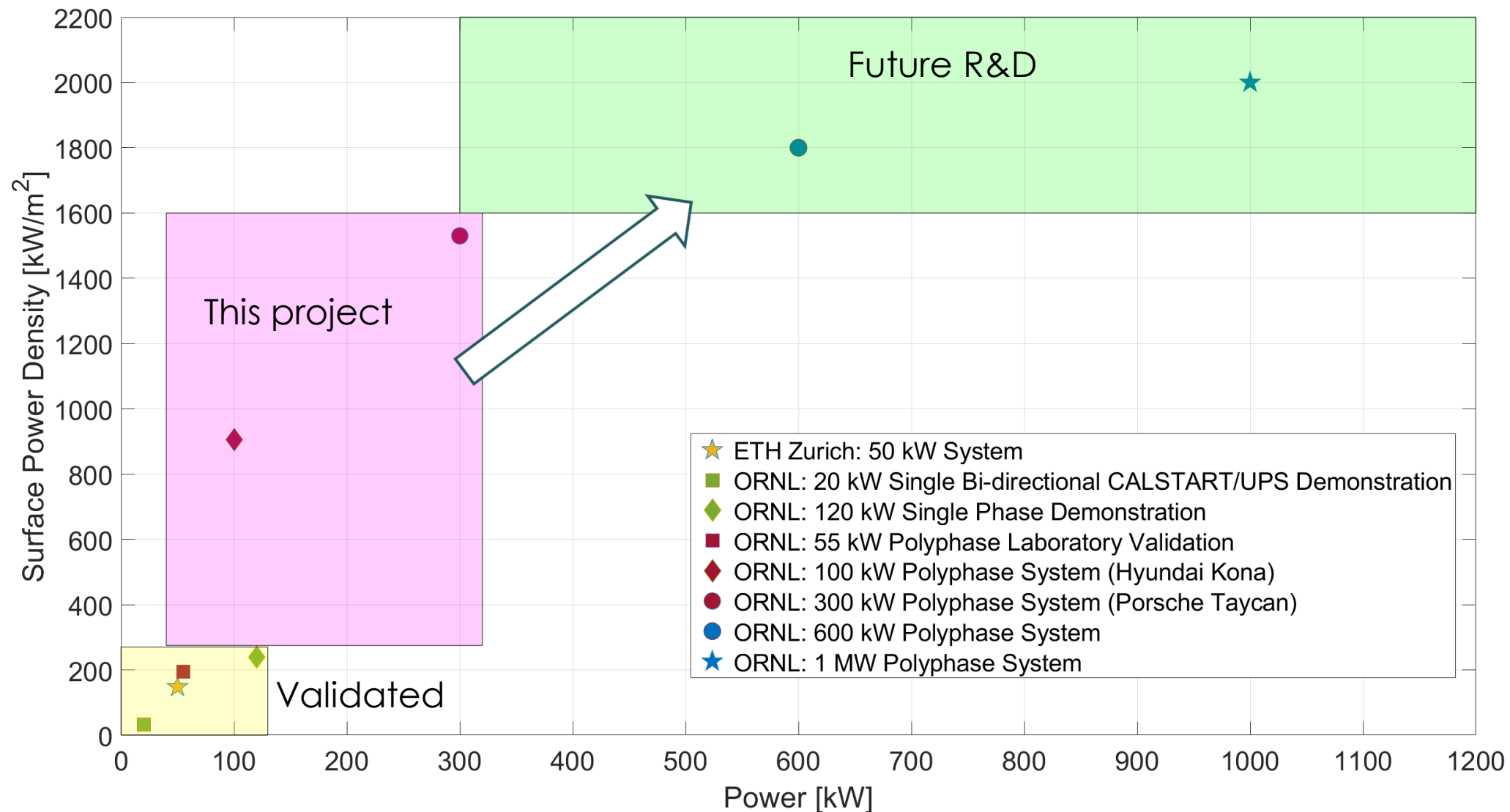
- 97.5% inverter efficiency
- 98.1% rectifier efficiency
- 99.49% coil-to-coil efficiency
- **95.12% overall efficiency**

dc input and dc output voltage,
current, and power measurements
for dc-to-dc efficiency analysis



Technical Accomplishments – BP I

Comparisons with the state-of-the-art systems

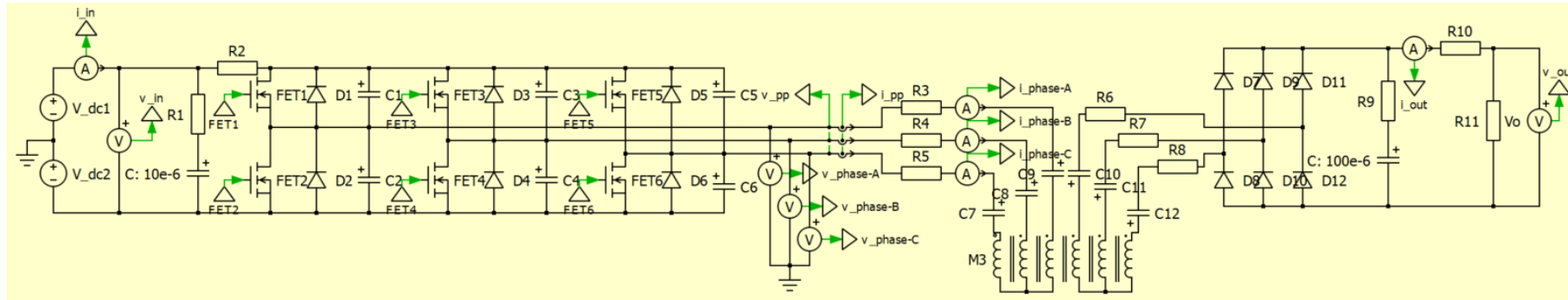


Power transfer level and surface power density of state-of-the-art stationary wireless power transfer systems

Technical Accomplishments – BP II

Developed Simulations and Detailed Loss Models for Power Stage Development Considerations

- Created a simulation model using the designed polyphase coupler parameters for comparisons of the inverter power modules (CREE and ROHM SiC modules).
- Evaluated a single 3-phase inverter for up to 300 kW in simulations.



Simulation model of the power stage for power device evaluations.



CREE SiC Module

*CAS325M12HM2
(1200V/356A)*



ROHM SiC Module

*BSM600D12P3G001
(1200V/600A)*

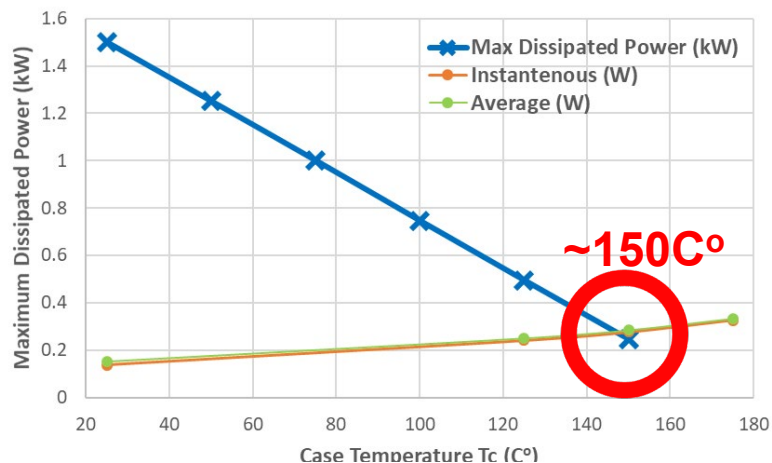
- Determined the voltage and current required for 300 kW
- Developed detailed loss models based on the device characteristics considering the turn-on time, turn-off time, dead-time, and diode reverse recovery time
- Analyzed thermal performance to evaluate these devices in a 3-phase inverter arrangement

Technical Accomplishments – BP II

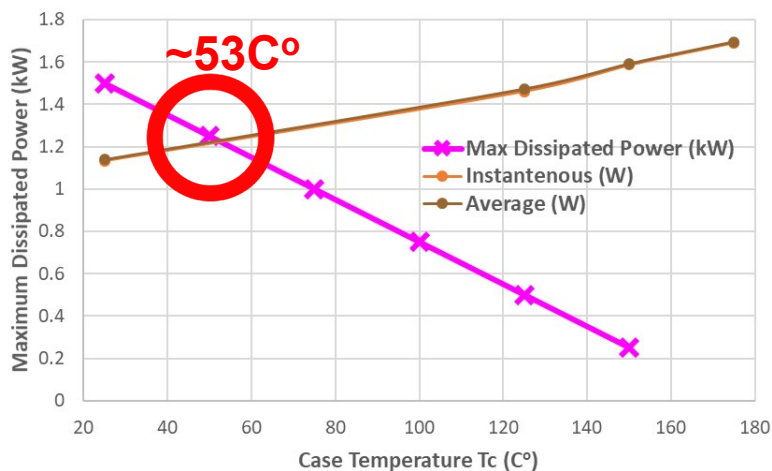
Completed the Thermal Analysis for Power Modules for the Feasibility of 3-Phase Inverter at 300 kW

CREE SiC Module CAS325M12HM2

Primary-FET Power Dissipation vs Case Temp. $T_j \leq 175^\circ\text{C}$

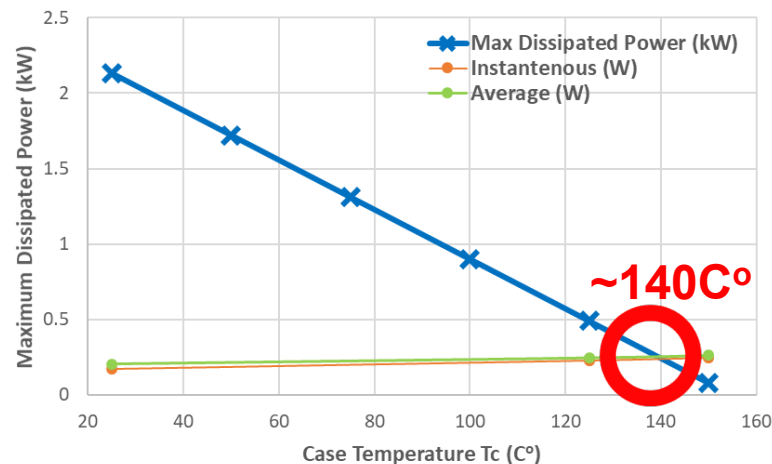


Secondary-FET Body Diode Power Dissipation $T_j \leq 175^\circ\text{C}$

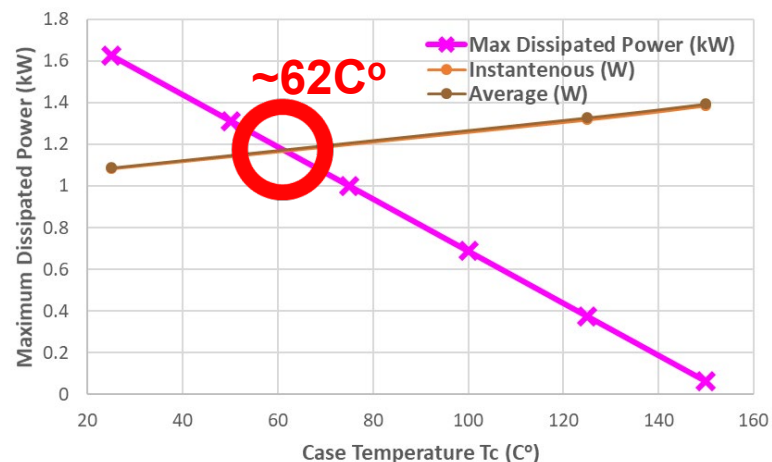


ROHM SiC Module BSM600D12P3G001

Primary-FET Power Dissipated vs Case Temp. $T_j \leq 175^\circ\text{C}$



Secondary-FET Body Diode Power Dissipation $T_j \leq 175^\circ\text{C}$

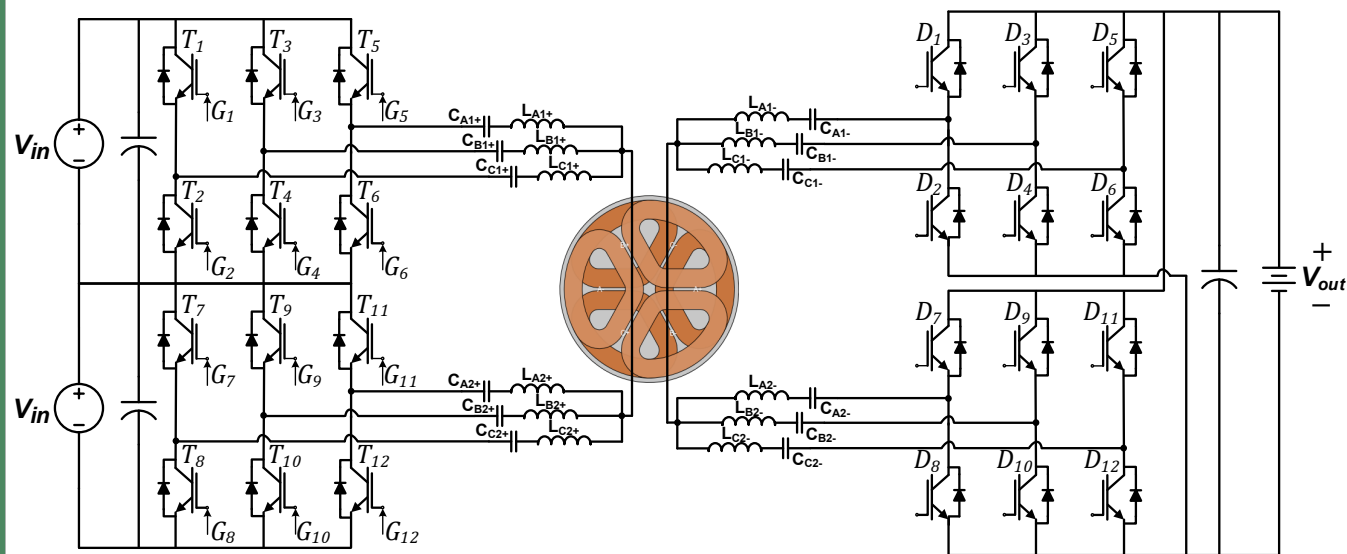


- Both power modules can be used to form a single 3-phase inverter for up to 300 kW operation but they will operate very close to their thermal limits
- Long-duration operation may not be guaranteed
- Lifetime may be an issue for long-term if operated at these points
- These modules could be safely used on the vehicle side rectifier but it would be a very expensive solution for the vehicle side
- 2 parallel inverters or more advanced designs would be needed for reasonable thermal conditions

Technical Accomplishments – BP II

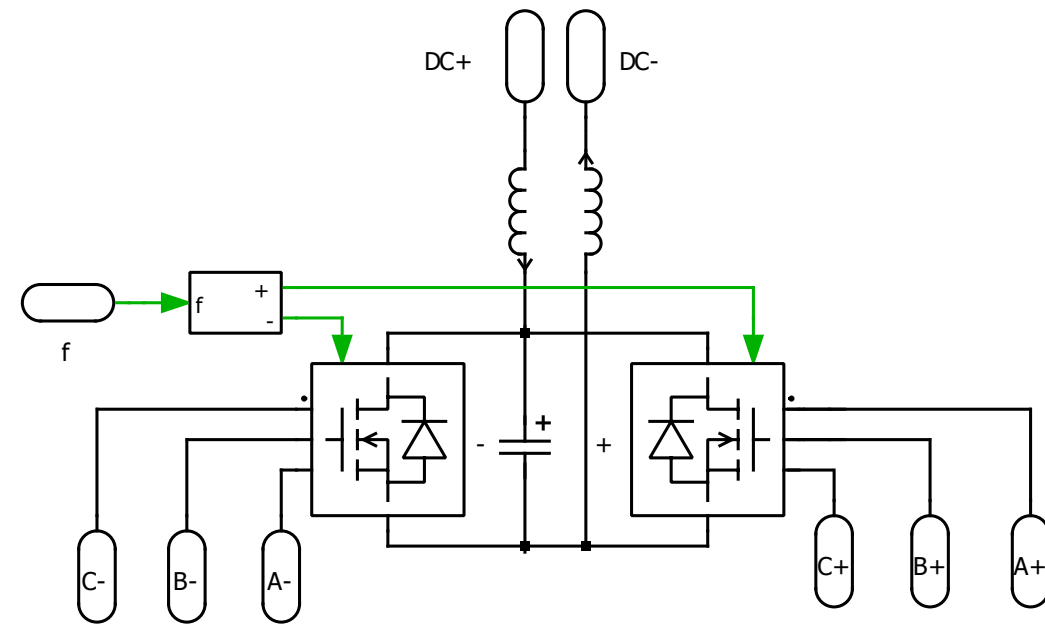
Analyzed Advanced Inverter Architectures for the Primary-Side

Series Cascaded Inverter (Multilevel inverter)



- Would require dividing the input power blocks into two and synchronizing them
- Would also require two high-precision input capacitors and a more complicated dc bus bar design
- More complicated controls
- More relaxed voltage limitations, higher efficiency

Open-ended winding dual inverter topology



- Open-ended winding/six-phase design
- Each inverter sees the same dc bus voltage
- Two three-phase inverters with shared dc-Link capacitors
- Doubles effective output voltage while maintaining dc-link capacitor size reduction compared to 3 single-phase H-bridges

Technical Accomplishments – BP II

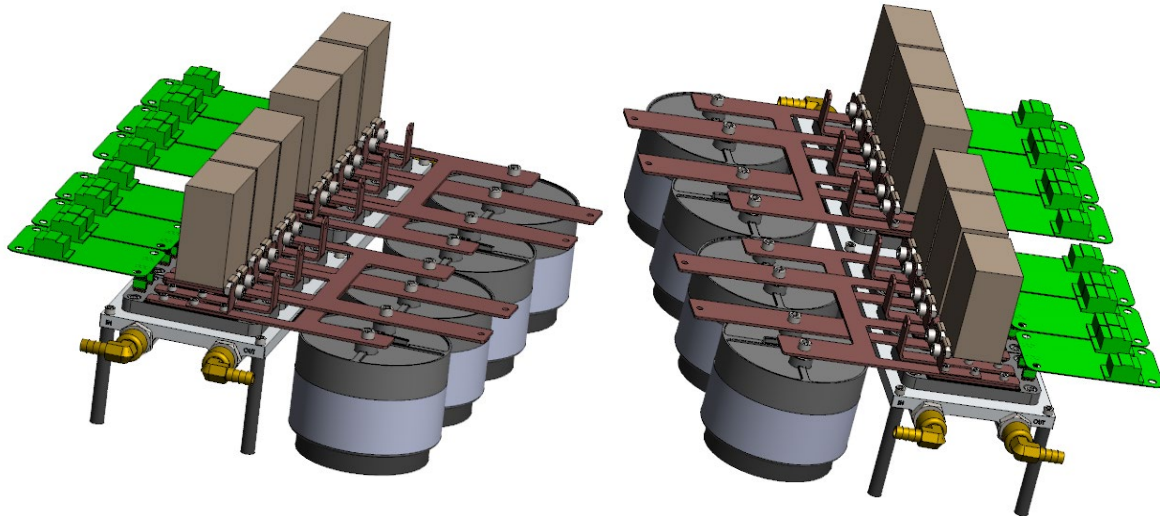
Compared the Power Stage Architectures and Determined the Inverter Design

- Open-ended dual inverter, 6-phase parallel (two 3-phase inverters simply paralleled), and 3-phase multi-level converter topologies reduce the primary current in the SiC FET switches.
- Primary coil current is reduced half in 6-phase parallel connection and almost four times less in open-ended dual inverter and 3-phase multi-level inverter options.
- Also, secondary coil currents and secondary rectifier current RMS values are half for both 6-phase parallel and multi-level connection compared to the 3-phase connection. For open-ended dual inverter option, it is almost 5 times less considering 3-phase inverter.
- Considering the secondary side current reduction performance, 9-phase coil design and open-ended dual inverter option is selected.

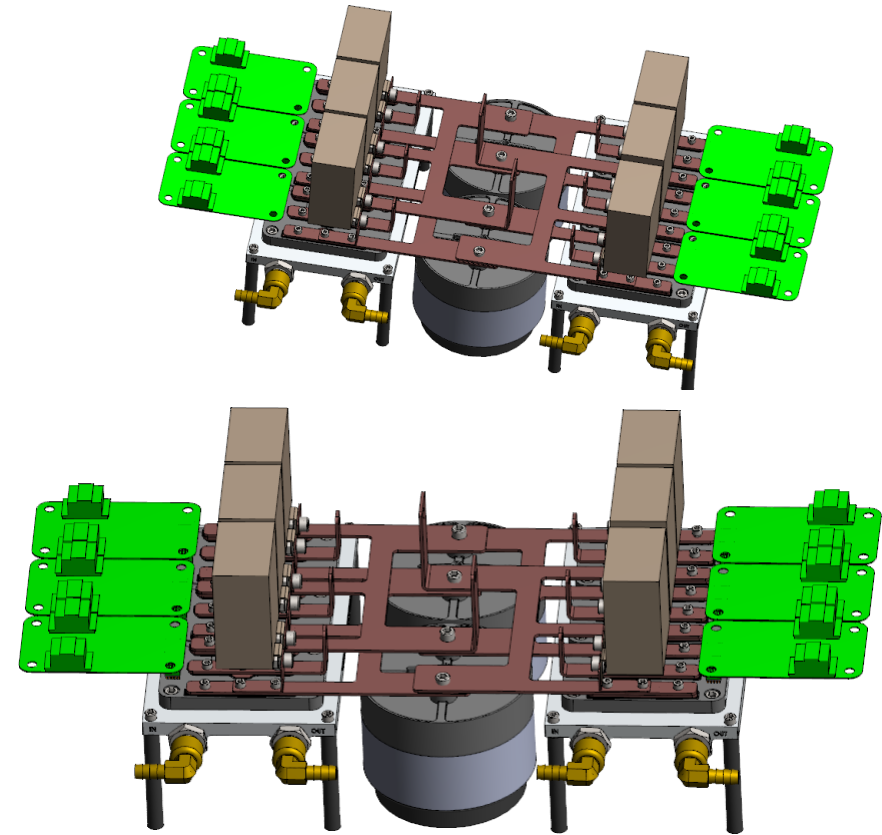
Inverter type	Phase-A current [A _{rms}]	FET current [A _{rms}]	Primary coil current [A _{rms}]	Secondary coil current [A _{rms}]	Secondary diode voltage [V _{rms}]	Secondary diode current [A _{rms}]
Single 3-phase inverter	291.4	204.8	289.3	553.6	288.24	394.6
6-phase conventional parallel inverter	144.3	100.4	144.2	277.6	285.6	196.2
Series cascaded multilevel inverter	71.8	101.4	71.7	277.7	285.4	197.1
Open-ended winding dual inverter	155.2	109.9	63.1	108.2	255.8	72.1

Technical Accomplishments – BP II

Analyzed Power Stage Design Alternatives for the Open-Ended Dual Inverter Configuration



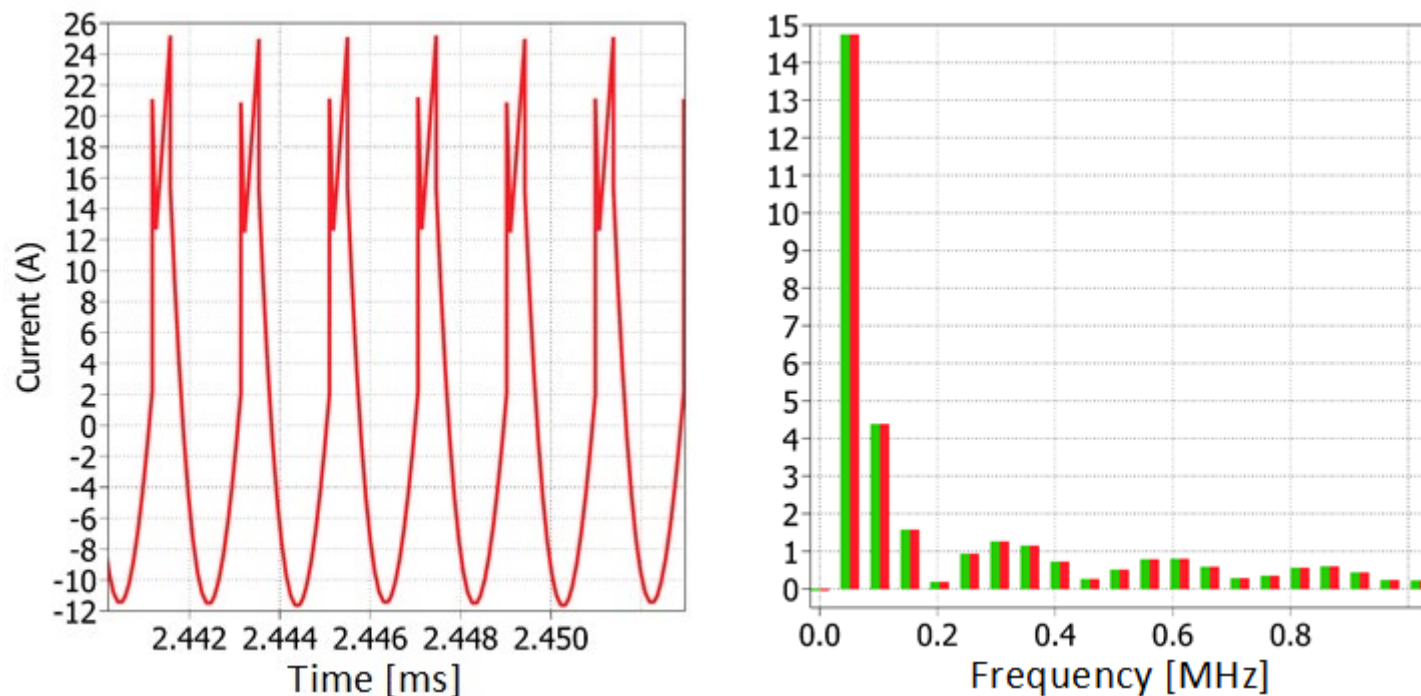
- Extended dc bus to connect two three-phase inverters
- Two three-phase inverters
- Simple but difficult to reduce volume by eliminating the capacitors
- Dimensions: 556mm (L) × 196 mm (H) × 341 mm (D)
- Volume: 37.16 liters
- Power density: 13.45 kW/liter



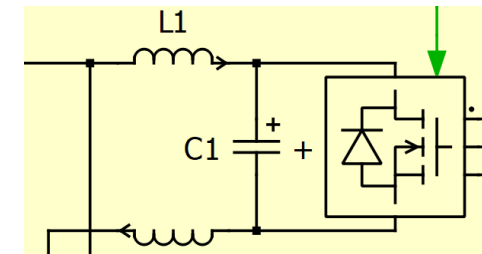
- Modified dc bus for connecting two three-phase inverters
- Reduced capacitors by half, more volume reductions possible
- Dimensions: 567mm (L) × 196 mm (H) × 276 mm (D)
- Volume: 30.67 liters
- Power density: 16.30 kW/liter

Technical Accomplishments – BP II

Analyzed Ripple Current Characteristics for Capacitor Sizing for 300 kW Output Power



Inverter dc-bus capacitor ripple current and its harmonic spectrum



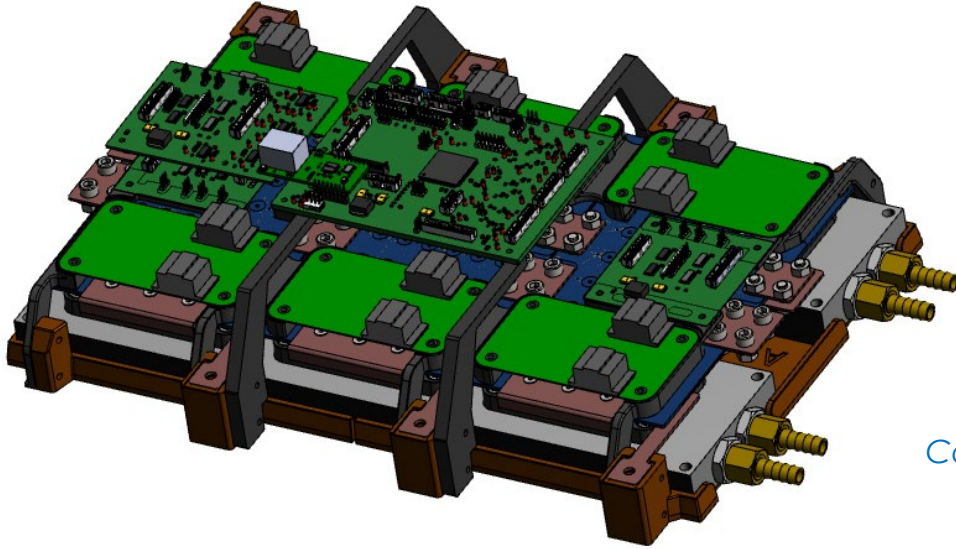
- Harmonics at 6th resonant frequency
- Ripple rms – 11.2 A for each of the 3-phase inverter
- Ripple frequency 510 kHz
- Results indicate very low current ripple harmonics and significant reductions in capacitance is possible
- Higher frequency capable capacitors are needed.

FFT Results for Current Requirement

Frequency	510kHz	1.02MHz	1.53MHz	2.04MHz	2.55 MHz	3.06MHz	3.57 MHz
Current amplitude	14.8	4.4	1.57	0.19	0.93	1.25	1.14

Technical Accomplishments – BP III

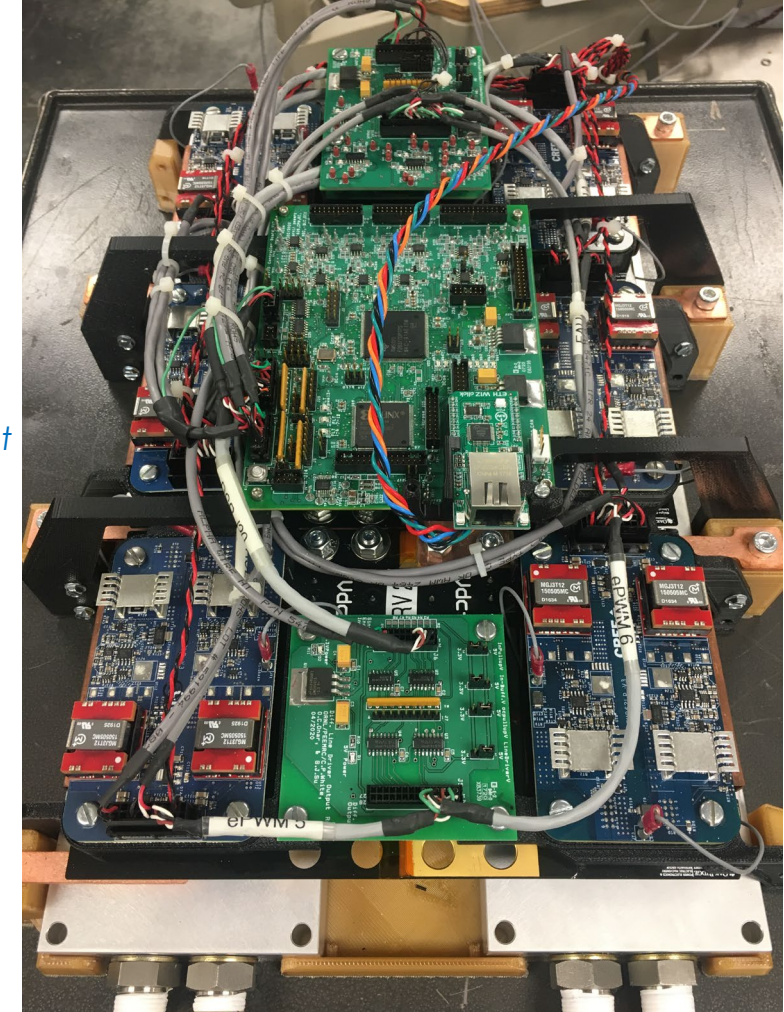
Completed the Open-Ended Dual Inverter Design with Two 3-Phase inverters with Custom Design Heatsink, Custom Design PCBs with TDK Ceralink® Capacitor Assembly



Open-ended winding dual inverter design with PCB dc bus bars with Ceralink capacitors

- Dimensions: 305.62mm (L) × 70.86 mm (H) × 423.22 mm (D)
- Volume: 9.16 liters
- Power density: **54.58 kW/liter**
- Custom designed PCB bus bars fabricated

Completed hardware development of the primary side inverter

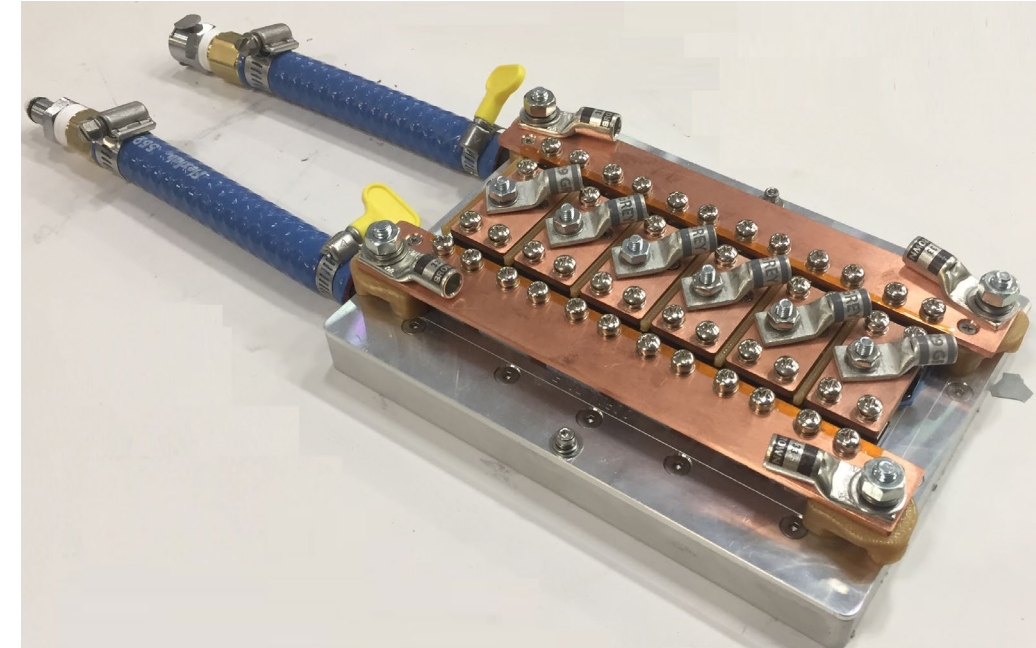


Technical Accomplishments – BP III

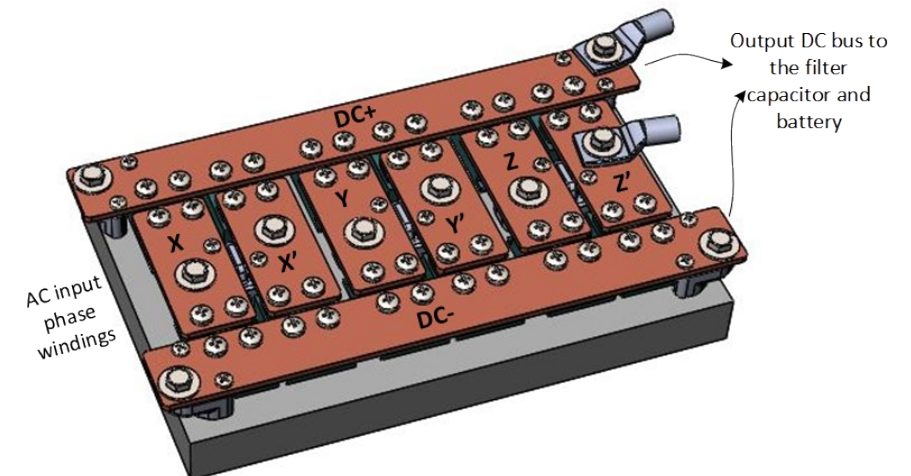
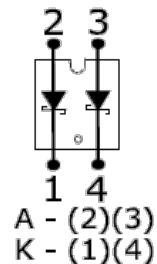
Analyzed the Vehicle-side Rectifier Design Considerations and Development

	GB2X100MPS12-227 SiC Schottky diode module T_j at 125°C	APTDF400U120G Si p-n junction module T_j at 100°C
Quantity	36	12
Total area of rectifiers	66.96 sq. in	284.76 sq. in
Losses	2686 W (0.89% of P_{out})	3060 W (1.02% of P_{out})
Cold plate requirements	< 0.032 C/W for all 8 modules on 1 single cold plate	= 0.0156 C/W for all 4 modules on 1 single cold plate
Price	\$135/module	\$65/module

100-kW open-ended winding dual Genesic SiC rectifier

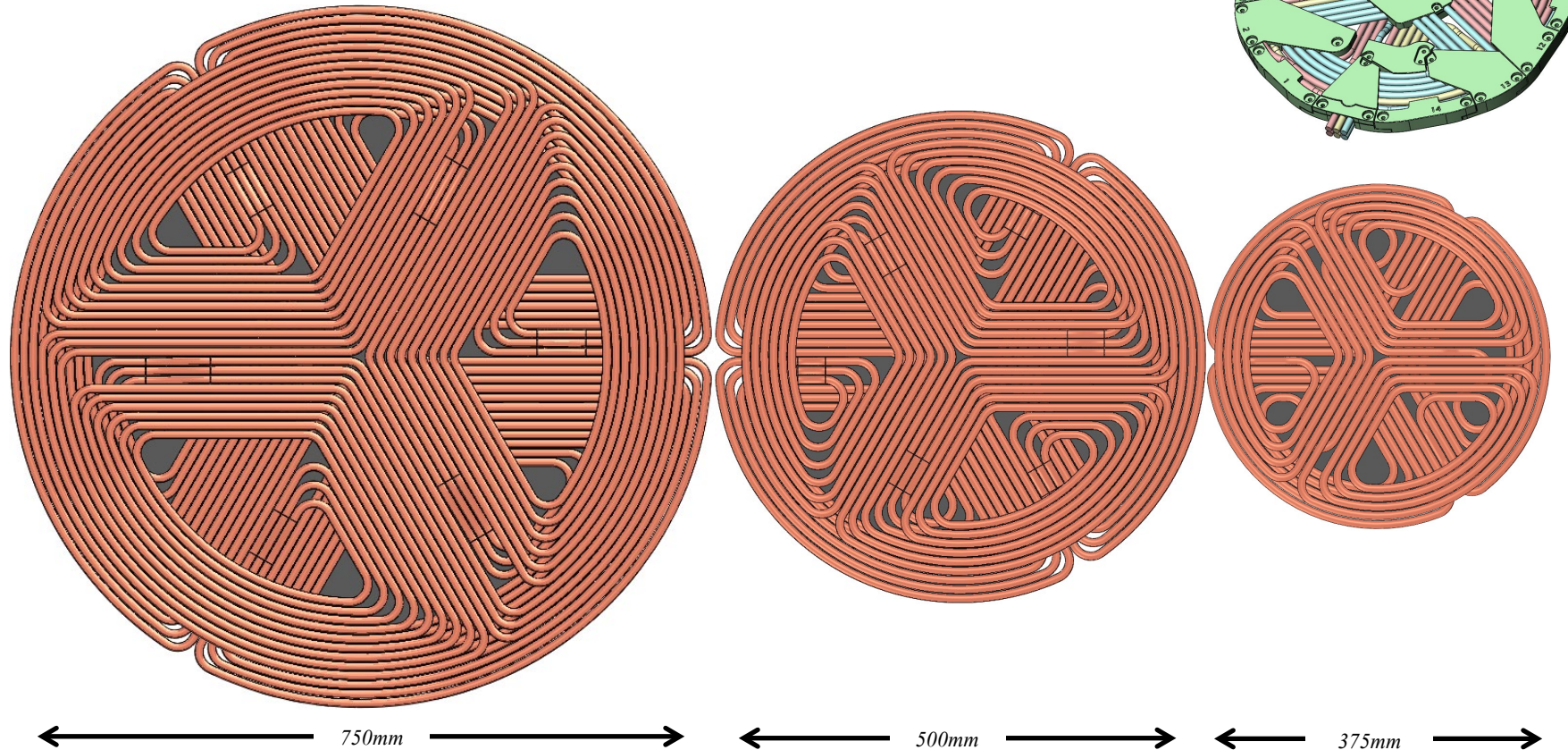


SOT-227 (Isolated Base)



Technical Accomplishments – BP III

Completed the Ground and Vehicle Coupler Designs and Developments



100kW/300kW Ground Coupler

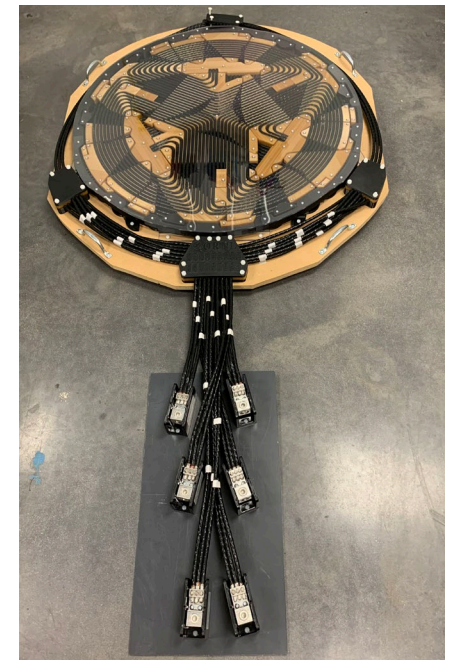
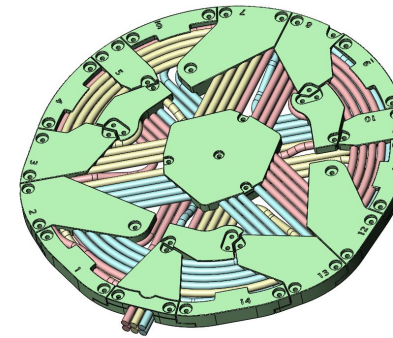
0.44 m²
0.68 MW/m²

300kW Vehicle Coupler

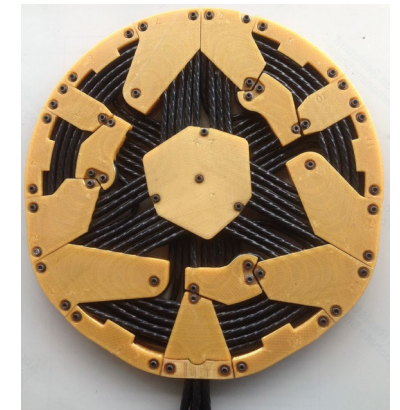
0.196 m²
1.53 MW/m²

100kW Vehicle Coupler

0.11 m²
0.9054 MW/m²



Completed laboratory development of the 300 kW ground coupler



Completed laboratory development of the 100 kW ground coupler

Technical Accomplishments – BP III

Completed Coupler Inductance Measurement and Characterization

- The 6-by-6 inductance matrix elements obtained by the FEA-based model

$$L_{design} = \begin{bmatrix} L_a & M_{ab} & M_{ac} & M_{ax} & M_{ay} & M_{az} \\ M_{ba} & L_b & M_{bc} & M_{bx} & M_{by} & M_{bz} \\ M_{ca} & M_{cb} & L_c & M_{cx} & M_{cy} & M_{cz} \\ M_{xa} & M_{xb} & M_{xc} & L_x & M_{xy} & M_{xz} \\ M_{ya} & M_{yb} & M_{yc} & M_{yx} & L_y & M_{yz} \\ M_{za} & M_{zb} & M_{zc} & M_{zx} & M_{zy} & L_z \end{bmatrix} = \begin{bmatrix} 14.5 & -4 & -3.99 & -2.39 & 1.03 & 1.04 \\ -4 & 14.5 & -4 & 1.04 & -2.38 & 1.03 \\ -3.99 & -4 & 14.5 & 1.03 & 1.04 & -2.38 \\ -2.39 & 1.04 & 1.03 & 14.6 & -3.95 & -3.96 \\ 1.03 & -2.38 & 1.04 & -3.95 & 14.6 & -3.96 \\ 1.04 & 1.03 & -2.38 & -3.96 & -3.96 & 14.6 \end{bmatrix} \mu H$$

- The 6-by-6 inductance matrix elements obtained by the FEA-based model

$$L_{measured} = \begin{bmatrix} 14.92 & -4.2 & -4.0725 & -1.4475 & 0.86 & 0.8 \\ -4.2 & 14.75 & -4.1225 & 0.8325 & -1.4475 & 0.915 \\ -4.0725 & -4.1225 & 14.9 & 0.7675 & 0.695 & -1.55 \\ -1.4475 & 0.8325 & 0.7675 & 16.08 & -3.93 & -3.477 \\ 0.86 & -1.4475 & 0.695 & -3.93 & 16.01 & -3.477 \\ 0.8 & 0.915 & -1.55 & -3.56 & -3.477 & 16.06 \end{bmatrix} \mu H$$

Technical Accomplishments – BP III

Finalized Coupler Specifications

	Interoperable Ground Assembly	100kW Vehicle Assembly	300kW Vehicle Assembly
Diameter	750 mm	375 mm	500 mm
Litz Wire	3×4AWG/phase	1×4AWG/phase	3×4AWG/phase
Litz + Ferrite Thickness	33.6 mm (18.6 + 15)*	28.6 mm (18.6 + 10)*	33.6 mm (18.6 + 15)*
Litz + Ferrite Mass	42.2 kg (9.9 + 32.3)*	8.8 kg (2.4 + 6.4)*	19.4 kg (4.7 + 14.7)*
Worst Case Losses	2362 W (697 + 1665)*	596 W (169 + 311)*	1343 W (331 + 1012)*
Surface area	0.44 m ²	0.196 m ²	0.11m ²
Surface power density	0.68 MW/m ²	1.53 MW/m ²	0.905.4 MW/m ²
Coil to Coil Efficiency		97.4%	98.8%

- Misalignment tolerance up to 125 mm in any direction for both receivers.

Technical Accomplishments – BP III

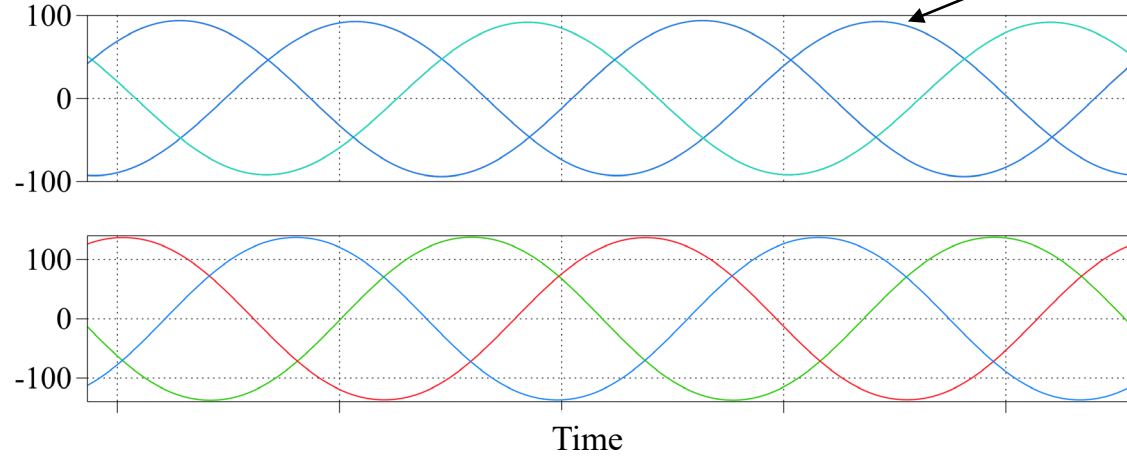
Analyzed the Component Stresses for Series and LCC Tuned Secondary-Side

Comparison	Secondary Tuning	Series	LCC
dc-link ripple current	Capacitor Size ($<10 V_{pp}$ ripple)	39 μF	15 μF
	Current ripple	155 A_{rms}	79 A_{rms}
Inverter output current [A]	Current rating	183 A_{rms} , 171 A_{avg}	159 A_{rms} , 140 A_{avg}
	Turn-off current	129 A_{rms} , 229 A_{avg}	18 A_{rms} , 110 A_{avg}
Primary coupler current [A]	Current rating	86 A_{rms}	88 A_{rms}
Secondary coupler current [A]	Current rating	142 A_{rms}	110 A_{rms}
Rectifier input current [A]	Current rating	381 A_{rms} , 344 A_{avg}	402 A_{rms} , 351 A_{avg}
Rectifier output current [A]	Capacitor Size ($<3.6 V_{pp}$ ripple)	175 μF	74 μF
	Current ripple	240 A_{rms}	100 A_{rms}

Technical Accomplishments – BP III

Completed Interoperability Simulations with Inverter, Resonant Network, and Transmitting Coil are Fixed for the Two Different Receivers

100kW

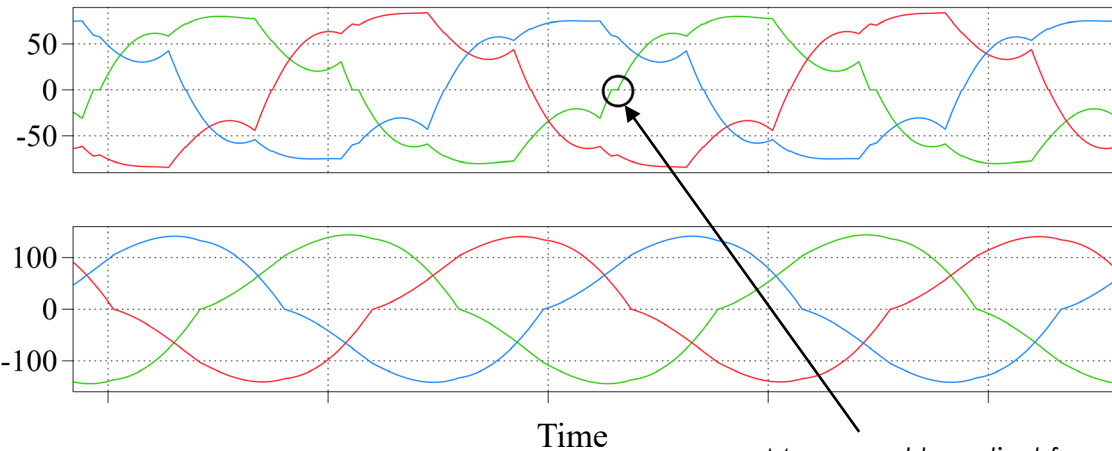
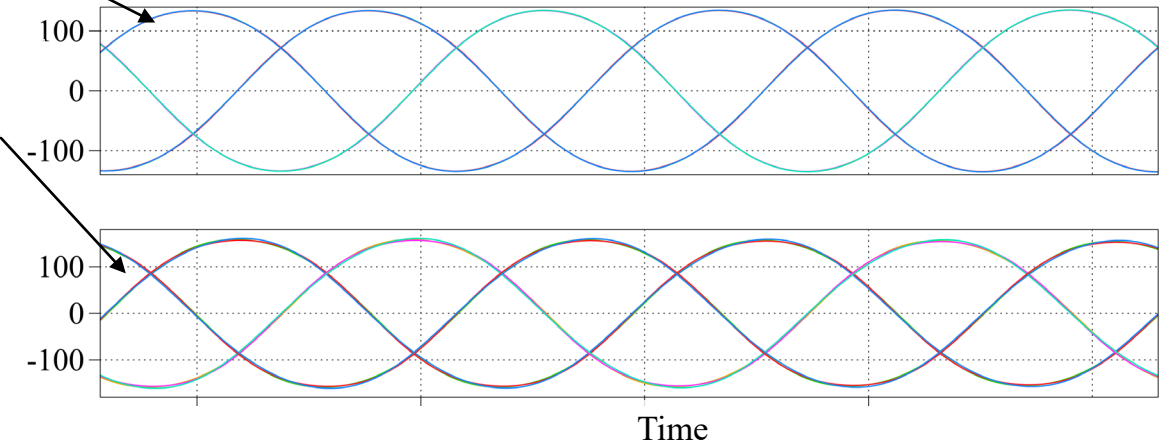


Currents balanced among 9 individual phase windings

Ground Coupler Currents

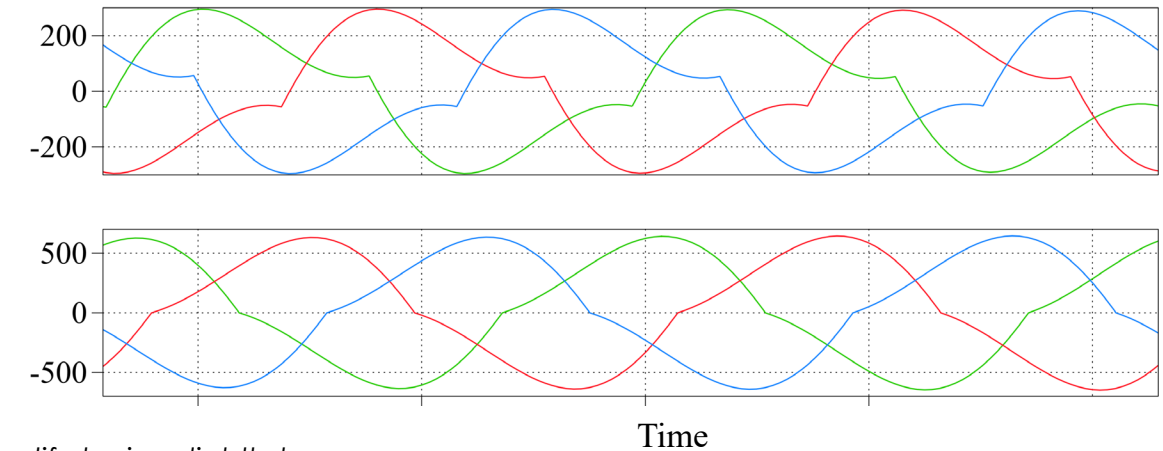
Vehicle Coupler Currents

300kW



Inverter Output Currents

Rectifier Input Currents



May need to adjust frequency or modify tuning slightly to improve inverter efficiency at lower power

Simulation results for two different receivers

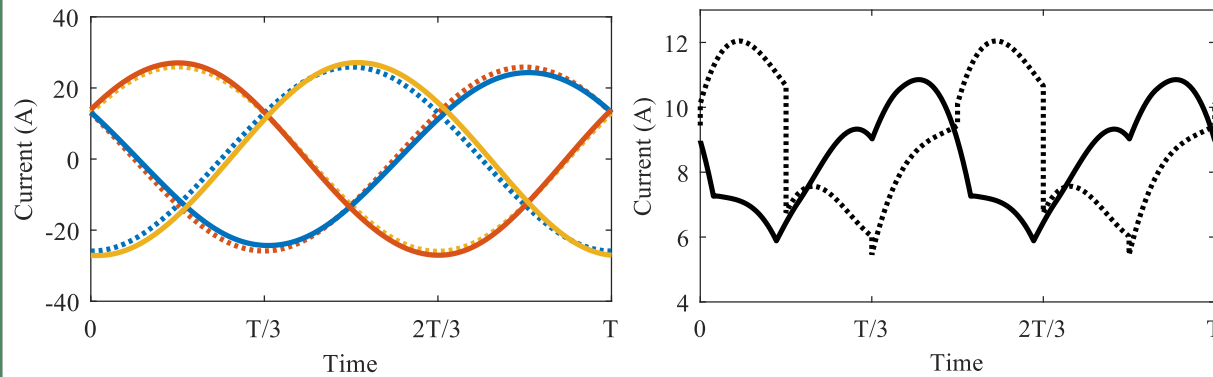
Technical Accomplishments – BP III

Analyzed Control Requirements for Three-Phase Inverter Control Under Misalignment

- 72% reduction in RMS output current ripple
- 90% reduction in 2nd output second harmonic
- Marginal impact on inverter input current ripple

Secondary Phase Current & Rectifier Output Current
 Primary Phase Current & Inverter Input Current

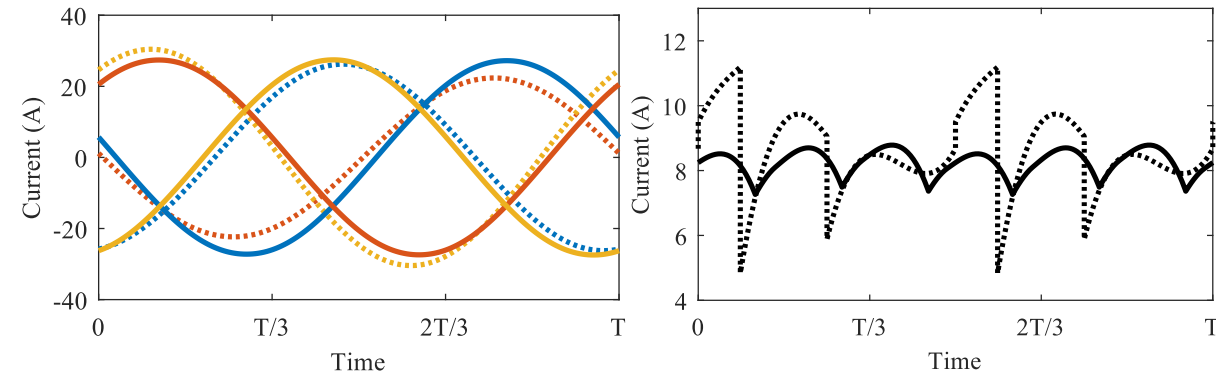
120 Degree Phase Shift



Primary and secondary coupler currents

Inverter input and rectifier output currents

Phase Shift for Low Output Ripple



Primary and secondary coupler currents

Inverter input and rectifier output currents

Mathematical Model:

$$\vec{I}_g = \frac{1}{j\omega} \mathbf{T}'_{\alpha\beta} (\mathbf{T}_{\alpha\beta} \mathbf{D}_{ll} \mathbf{M} \mathbf{T}'_{\alpha\beta})^{-1} \mathbf{T}_{\alpha\beta} \vec{V}_v$$

$$\vec{V}_g = \mathbf{D}_{ll} \left(j\omega \mathbf{L}_g + \frac{1}{j\omega C_{g,s}} \mathbf{I} \right) \vec{I}_g$$

$$\bar{V} = \frac{V_{bc}^g}{V_{ab}^g} = \frac{e^{j\phi_b} - e^{j\phi_c}}{1 - e^{j\phi_b}} = e^{j\frac{\phi_c}{2}} \frac{\sin(\phi_c/2 - \phi_b/2)}{\sin \phi_b/2}$$

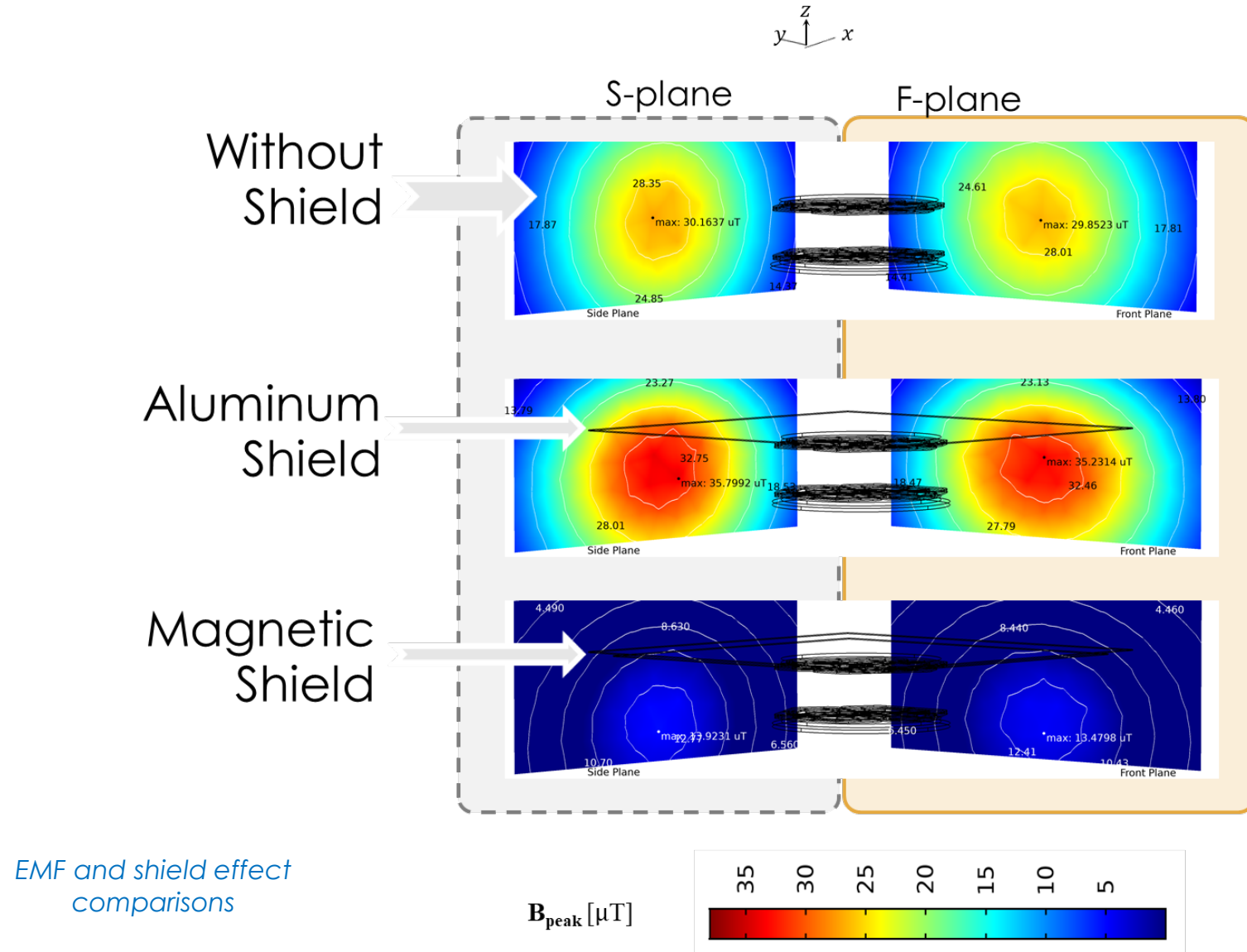
$$\phi_c = 2\angle \bar{V},$$

$$\phi_b = 2 \arctan \left(\frac{\sin(\phi_c/2)}{|\bar{V}| + \cos(\phi_c/2)} \right)$$

Technical Accomplishments – BP III

Analyzed Electromagnetic Field Emissions at 800 mm Away from the Center and Evaluated Shield Designs

- **Without** any shield, the peak flux density is close to the ICNIRP limit ($38 \mu\text{T}_{\text{peak}}$)
- **Aluminum** shield slightly increases the MFE on both plane
- **Magnetic** shield significantly suppresses the MFE

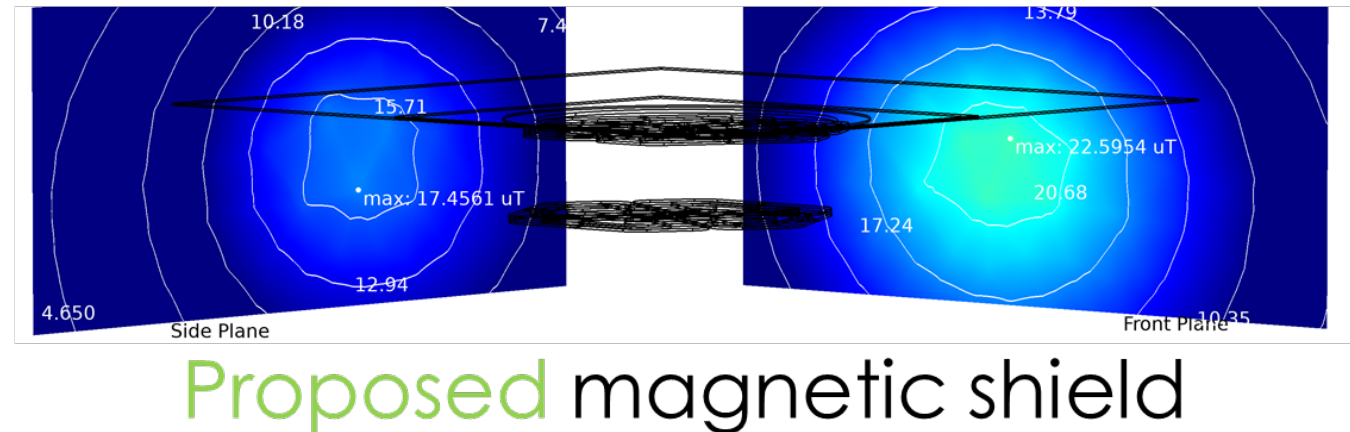
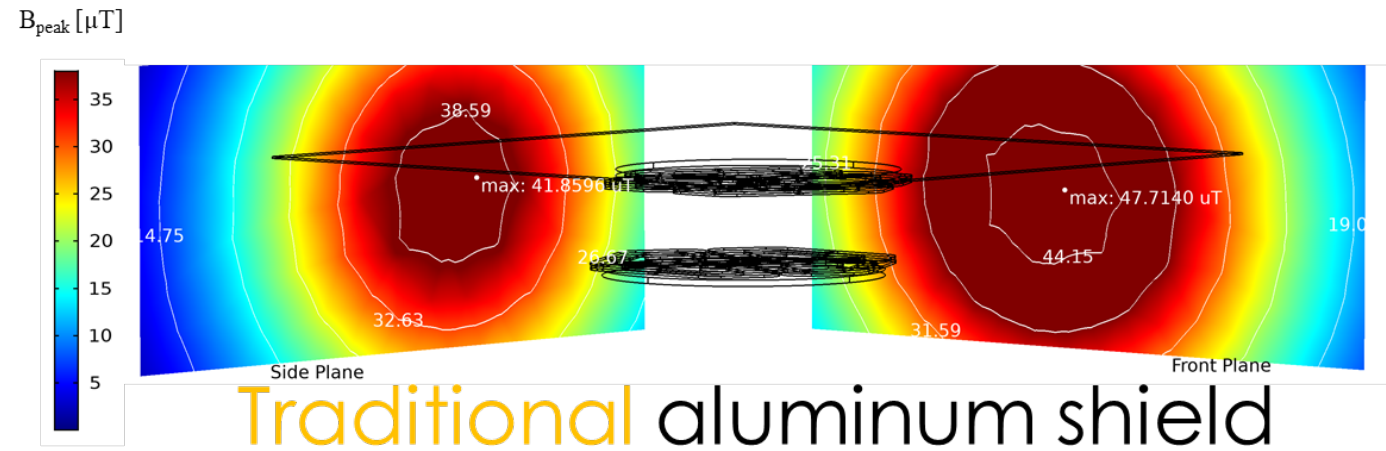
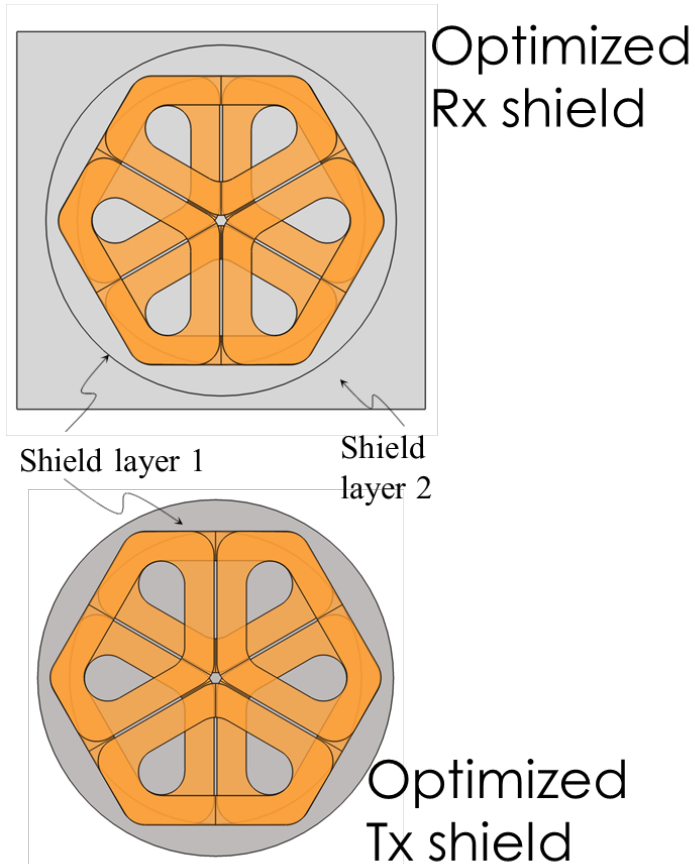


EMF and shield effect
comparisons

Technical Accomplishments – BP III

Optimized Magnetic Shielding and Results Comparisons at Misaligned Conditions

- X-misalignment (along the front)=75 mm
- Y-misalignment (along the side)=-100 mm

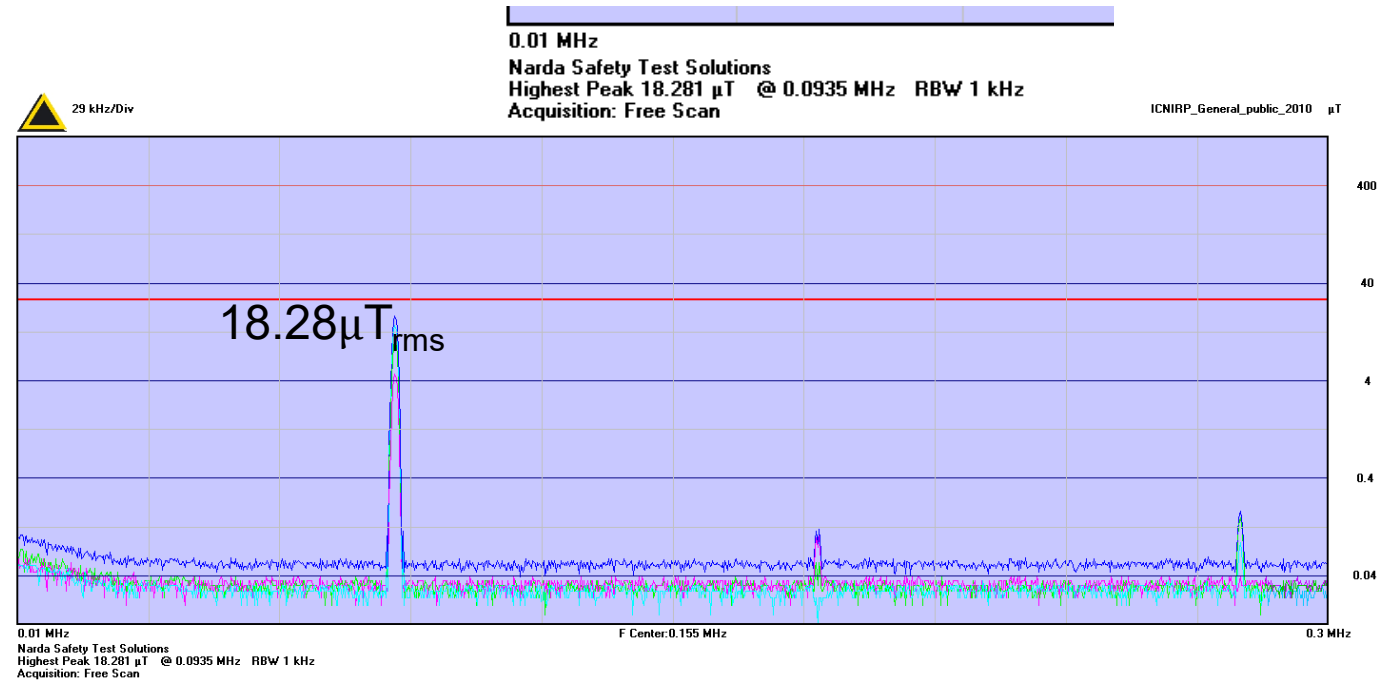
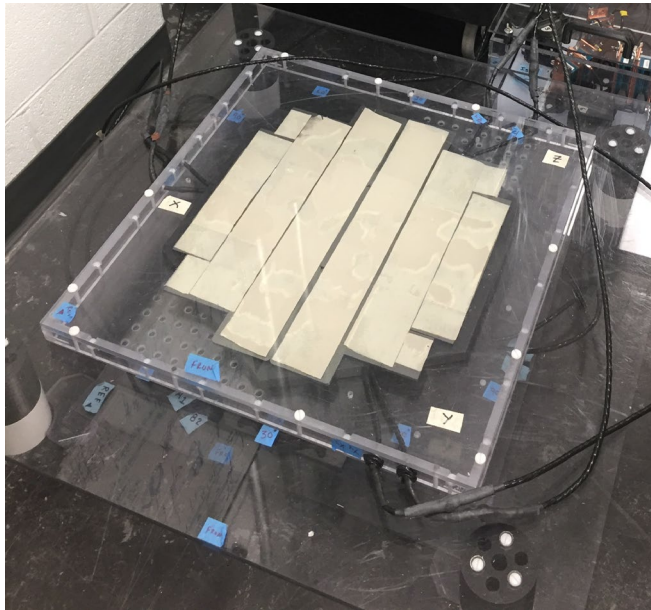
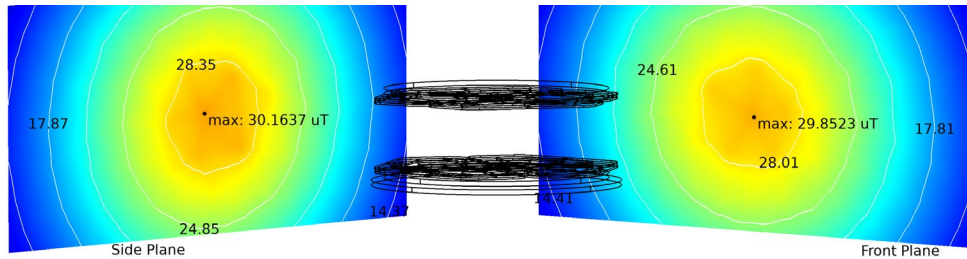


Optimized shield performance under misaligned conditions

Technical Accomplishments – BP III

Optimized Magnetic Shielding and Results Comparisons at Misaligned Conditions

- Three-phase polyphase coupler generates much lower MFE than the single-phase couplers

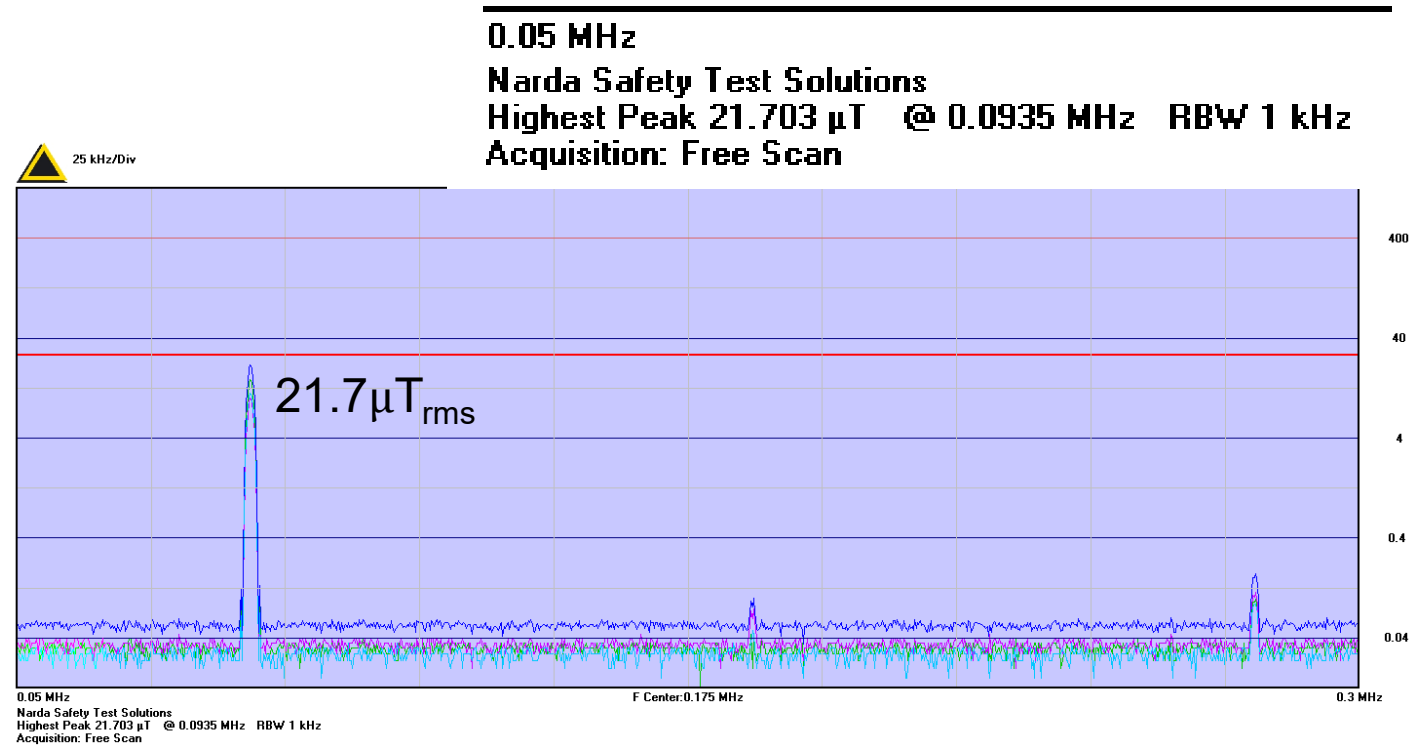


50 kW testing B-field without shield

Technical Accomplishments – BP III

MFE using Large Aluminum Shield

- Aluminum shield increases the MFE by 18%: from $18.28 \mu\text{T}_{\text{rms}}$ to $21.7 \mu\text{T}_{\text{rms}}$

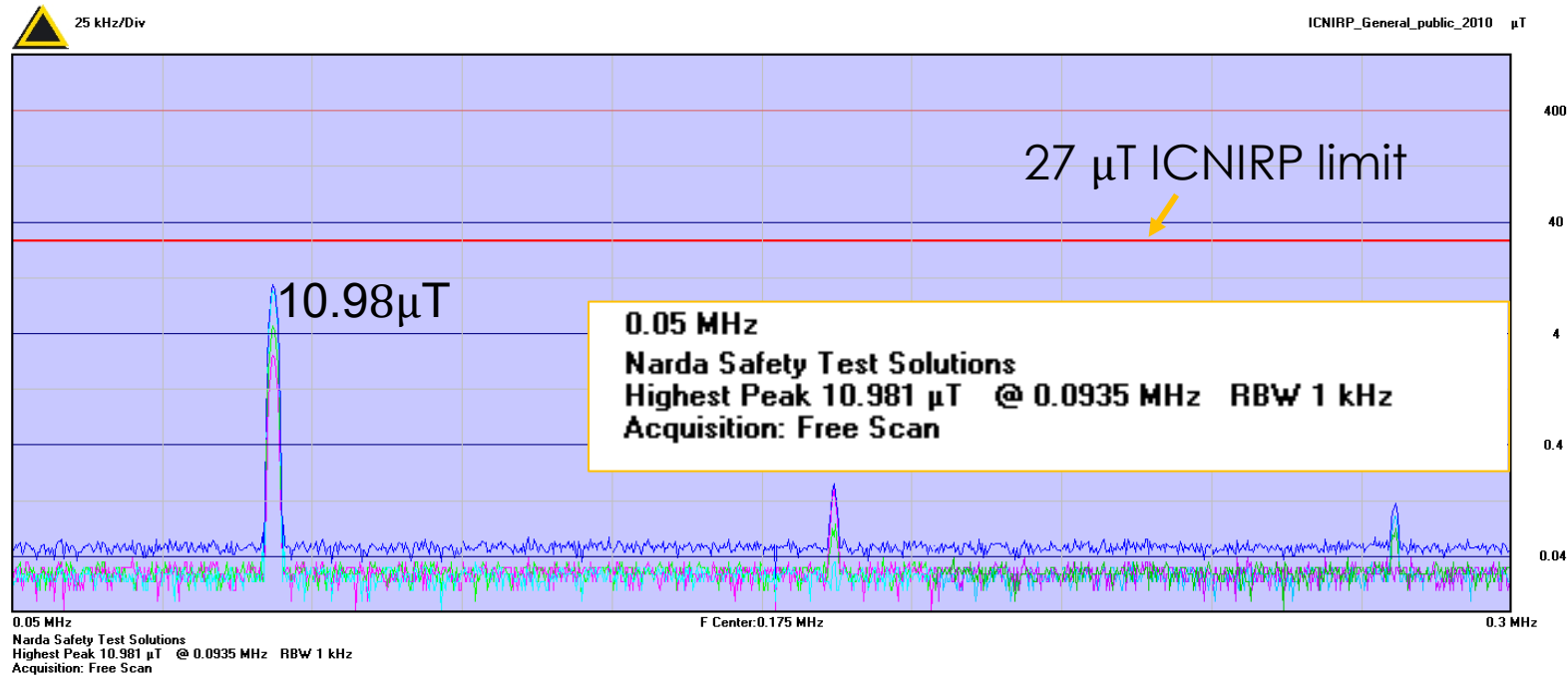


50 kW testing B-field with Al shield only

Technical Accomplishments – BP III

MFE using Magment and Aluminum-based Shield

- The MFE is reduced to $10.9\mu\text{T}$ (by 40%) using Magment and aluminum-based shield
- With magnetic & metallic shielding on secondary $<4.5\mu\text{T}$ field emission @0.98m away from primary @270kW



50 kW testing B-field with Magment and Al-based shield

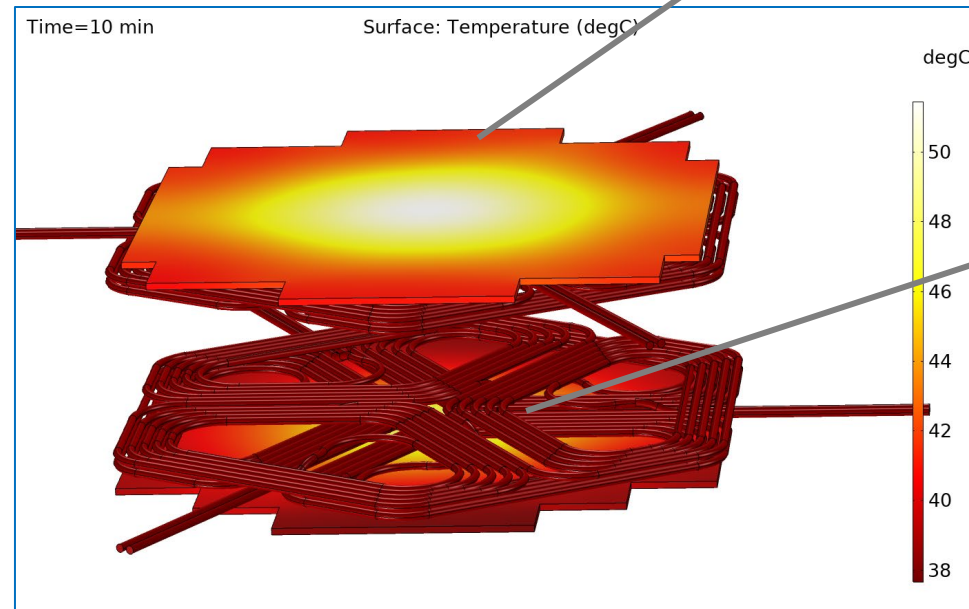
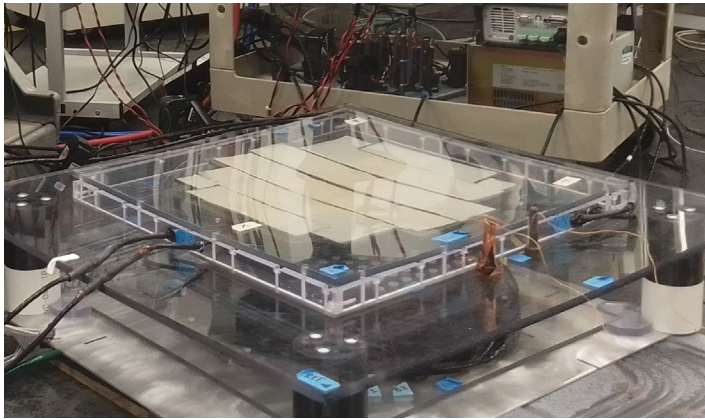
Technical Accomplishments – BP III

Thermal Analysis

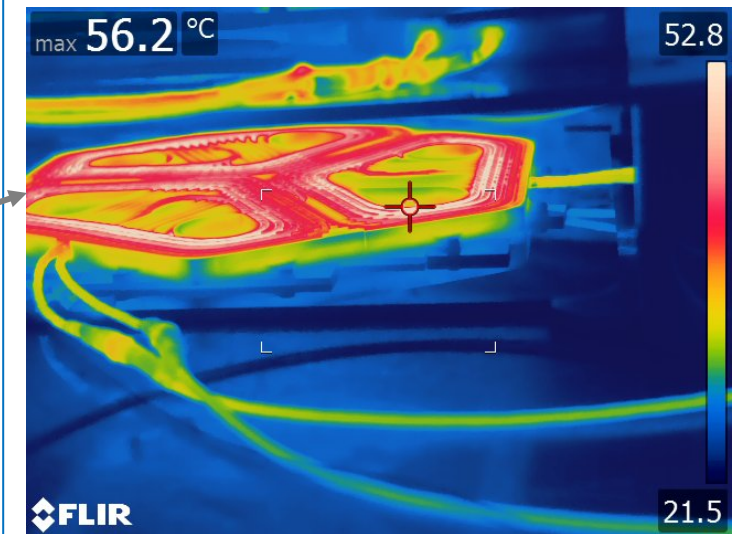
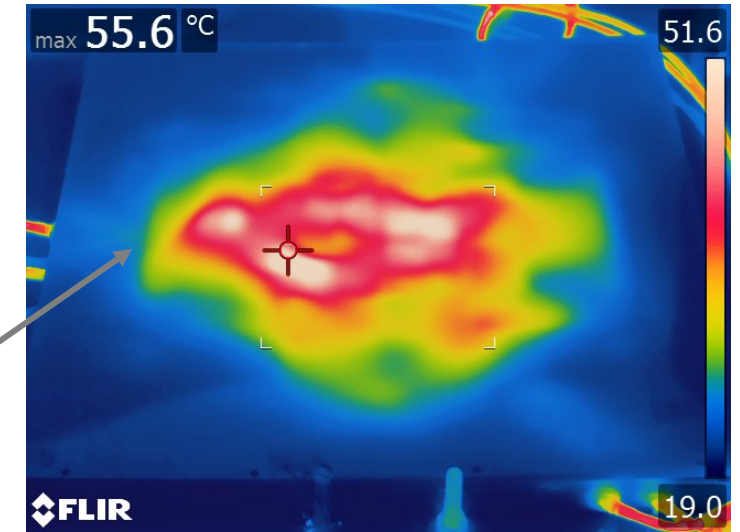
- Receiver is in a lexan enclosure
- Coupler pads will be filled with thermally conductive epoxy for increased heat dissipation and convection

Temperature readings at the end of the charging process; temperatures stay below 80°C

300W losses in each core

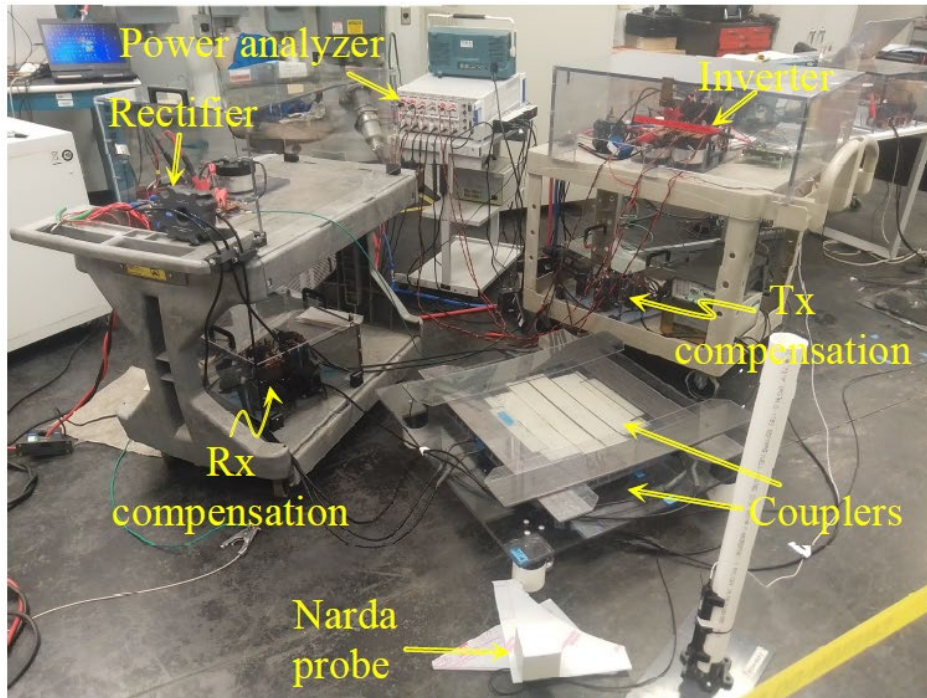


FEA simulation results

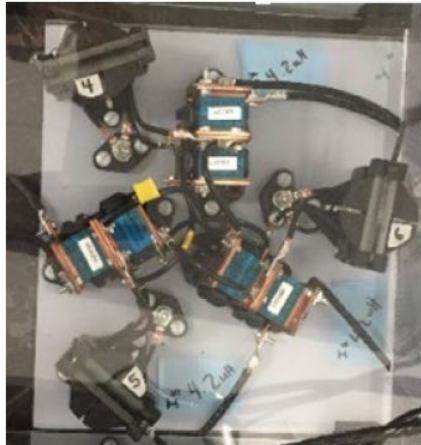


Experimental results

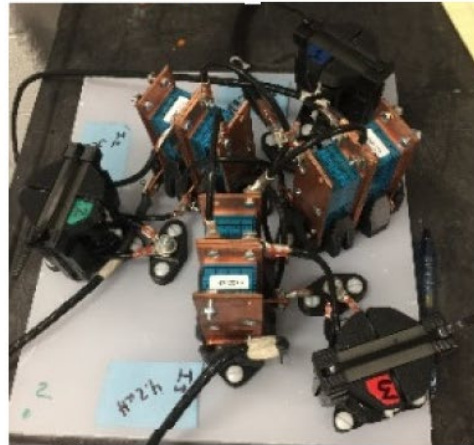
Bench Top Tests (ongoing)



(a) experimental setup

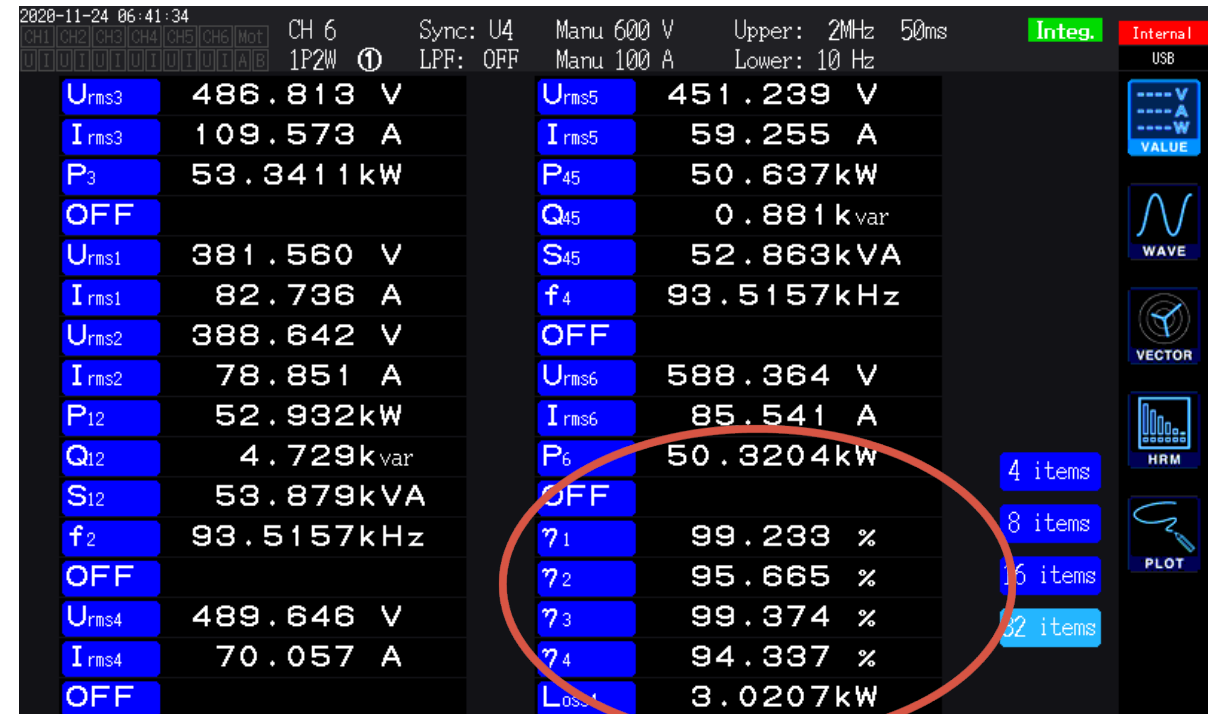


(b) Tx compensation



(c) Rx compensation

- Experimental validation of the double-sided LCC tuned polyphase WPT system continuous operation
- Also validated the 50 kW rectifier operation (@91 kHz)
- 50.3 kW output power, 99.23% inverter efficiency, 99.37% rectifier efficiency, 94.34% overall efficiency



Response to Previous Year Reviewers' Comments

- Project team would like to thank all of the reviewers for the feedback and the comments. All of the comments were positive and encouraging.
- Some of the comments:
 - *The approach is solid with design, testing, modeling, and validation.*
 - *The project is focused on both 100 kW and 300 kW inductive charging system, designed to be small with electromagnetic coupling. The team seems to have a very specific and detailed approach very well aligned with accomplishing the necessary activities to achieve project goals. In particular, the team has worked to take an approach to overcome shortcomings of existing systems. They have also included analysis of design alternatives, in order to minimize risk.*
 - *The approach is logically organized. Initially doing the soft requirements like iterative design and modeling. Later in the project there is demonstration of the 100 kW and the 300 kW systems on actual vehicles.*
 - *Overall, the project has appeared to achieve significant accomplishments. So far, they are showing 95% efficiency. The team compared their system against state-of-the-art systems, showing much higher power transfer at a larger air gap and at higher efficiency.*
 - *Good accomplishments occurred including the design and prototype, simulation and analysis of the components, and inverter design.*
 - *Collaboration was comprised of the National Laboratories, a charging provider, and two automotive companies. Each team member has a distinct piece of the project that compliments the other components.*
 - *The future work described seems like the next logical steps with this effort. Two areas for future consideration are interoperability with SAE Standard J2954-2, and evaluation of system costs for commercial implementation.*
 - *Will this technology be interoperable with the technology that is occurring with SAE J-2954-2?*
 - *Yes, through the primary-side inverter controls, polyphase couplers can be operated mimicking a circular coupler or a DD coupler and it is interoperable with these types of receivers.*

Collaboration and Coordination with Other Institutions

- **ORNL:** Project lead, project management, budget management, reporting, overall coordination, design, development, integration, testing, demonstration. Designer and developer of all the power electronics and electromagnetics components and subsystems.
- **Hyundai-Kia:** Vehicle manufacturer; providing the research vehicle along with engineering and integration support, support on BMS, battery thermal management, and technical specifications for the 100kW tests and integrations.
- **VW Group of America:** Vehicle manufacturer; providing the research vehicle along with engineering and integration support, support on BMS, battery thermal management, and technical specifications for the 270-kW tests and integrations.



Remaining Challenges and Barriers

- The electric and electromagnetic field emissions should be less than the limit levels set by the ICNIRP 2010 guidelines inside and around the vehicle while transferring 100 and 300 kW across 6-7 inches airgap.
- The thermal management system of the hardware requires utilizing cooling system designed for the vehicle's on-board electronics (traction drive inverter or on-board charger). Controlling the cooling system as well as other on-vehicle auxiliary components including the BMS system requires input from the OEM partners.

Any proposed future work is subject to change based on funding levels

Proposed Future Research

- **FY 2021**

- Integrate the final design to a Hyundai-Kia vehicle (Kona) with 100kW power transfer demonstration

- **FY 2021**

- Integrate the final design to a Porsche Taycan EV with 270-kW power transfer demonstration (~96 kWh battery pack)

Any proposed future work is subject to change based on funding levels

Summary

- **Relevance:** Increase the benefits and reduce the barriers in vehicle electrification, wireless charging, and vehicle to grid integration, achieve extreme fast charging power levels with advanced and compact designs with user convenience.
- **Approach:** Proposed a polyphase electromagnetic coupler and high-power and high-frequency power electronic converters to meet high-power and high-efficiency targets.
- **Technical Accomplishments:**
 - Designed and developed a 55 kW proof of concept prototype for laboratory tests and evaluations to scale the system and provide a design guideline for 100 kW and 270 kW power levels.
 - Currently testing the whole system together at different operating conditions including x-y and rotational misalignments.
- **Collaborations and Coordination with Other Institutions:**
 - **ORNL:** Project lead and project management.
 - **Hyundai-Kia:** Vehicle OEM. Providing a test vehicle and integration and engineering support and guidance.
 - **VW Group of America:** Vehicle OEM. Providing a test vehicle and integration and engineering support and guidance.
- **Future Work:**
 - Perform vehicle integrations and prepare for demonstrations for 100- and 270-kW power levels on two different research vehicles.

Any proposed future work is subject to change based on funding levels